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MILITARY SPECIFICATION
VEHICLE, COMBAT ENGINEER FULL TRACKED: 165MM GUN, M728

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of full-tracked armored combat vehicle which mounts a 165MM gun as primary weapon.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-527	- Enamel, Alkyd, Lusterless

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2350

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MIL-M-45212	- Mount, Gun; Mount, Howitzer.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-V-62060	- Vehicle, Combat Engineer, Full-Tracked: 165MM Gun, M728; Processing for Storage and Shipment of.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-193	- Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210	- Climatic Extremes, for Military Equipment.
MIL-STD-417	- Classification System and Tests for Solid Elastomeric Materials.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

8736334	- Vehicle, Combat, Engineer Full Tracked 165MM Gun, M728.
8752278	- Force Gage.
10940104	- Installation, Mount, Combination Gun, M150.
10940135	- Production Lubrication Chart.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a preproduction vehicle of the complete vehicle to be furnished under this specification, shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 First production vehicle. A first production vehicle shall be fully representative of vehicles to be furnished under the contract and shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components are free from defects that compromise, limit, or reduce the vehicle or component capabilities. Vehicles made in accordance with this specification shall be subjected to examinations and tests to prove vehicle capabilities to perform in accordance with the following:

a. Vehicle shall require only organizational maintenance (first and second echelon) during 2,000 miles of normal operation.
b. Vehicle shall require no replacement or major overhaul of any major automotive component during 4,000 miles of normal operation.
c. Complete track assemblies or individual shoes shall not require replacement during 2,000 miles of normal operation.

3.2.1.1 Vehicle operation. The 4,000 miles of vehicle operation combat loaded or simulated load that will not change the normal combat load center of gravity shall consist of the following:

a. 25 percent on paved roads either concrete, asphalt, or any combination of the two.
b. 25 percent on gravel and dirt roads with at least 10 percent of this distance under mud conditions.
c. 25 percent on level cross country.
d. 25 percent on hilly cross country.

3.3 Materials.

3.3.1 Qualified products. All assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4 and necessary in the construction of this vehicle, contractor shall be responsible for using only those products that are listed (by part or drawing number as QPL items) on the QPL in the referenced specification, or which have been approved for inclusion on such lists.

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Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made, or date of approval letter when products are approved but not as yet listed on the QPL.

3.3.2 Ozone resistance. Unless otherwise specified on detail drawings, all rubber components that are under tension or which may be flexed, shall be ozone resistant to the degree required by the "C_1" test requirements of MIL-STD-417.

3.4 Construction. Vehicles, components, subassemblies, and assemblies shall be fabricated and assemblies into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts List (EPL) 8736334. All parts requiring identification shall be marked in accordance with MIL-STD-130.

3.4.1 Sealing.

3.4.1.1 Sealer. The sealer shall prevent the entrance of water into the crew compartment when fording streams or rivers during cross country operation and when operating under water.

3.4.1.2 Seals. Seals shall prevent the entrance of all foreign matter with the exception of water as noted in 3.6.8.5 when fording shallow or under water. All seals shall prevent leaking of lubricants from bearings except those seals designed to allow the exit of flushing lubricant.

3.4.1.3 Hatch seals. Hatch seals shall prevent any leakage of water into the crew compartment.

3.4.2 Stowed equipment. All On Vehicle Equipment (OVE) items shall be capable of being stowed (combat loaded) in the spaces provided.

3.4.3 Cooling system. With vehicle operating in ambient temperature of 115° degrees Fahrenheit (F), the engine oil temperature shall not exceed 250°F at the heat exchanger outlet and the transmission oil temperature shall not exceed 300°F at the heat exchanger inlet.

3.4.4 Fuel system. During engine operation, the fuel system shall maintain fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when vehicle is being operated on 30 percent side slope with each side of vehicle up slope.

3.4.4.1 Fuel tanks and hydraulic oil reservoir. Provisions shall be made to assure internal cleanliness of the fuel tanks and all hydraulic oil reservoirs prior to initial fueling and filling. The fuel tanks shall be capable of receiving 50 gallons of fuel per minute.

3.4.4.2 Fuel and hydraulic lines. Provisions shall be made to assure internal cleanliness of fuel and hydraulic lines and connections prior to initial filling of systems. All fuel and hydraulic lines, connections, and tanks shall be free from leakage after installation and after break-in run.

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3.4.4.3 Heater fuel feed. With either of the fuel system pumps and the heater pump operating, the vehicle fuel system shall produce a minimum pressure of 7 psi at the outlet of the heater pump line.

3.4.5 Ventilation system. The ventilation system shall prevent harmful accumulation of Carbon Monoxide (CO) in the occupied areas of the vehicle. The concentrations of CO shall not exceed the percentage levels indicated below throughout time periods specified.

Maximum CO Concentration	Duration Period of CO Concentration
0.6 percentage	One minute or less
0.12 percentage	1 to 5 minutes
0.04 percentage	5 to 15 minutes
0.02 percentage	15 to 30 minutes
0.01 percentage	30 to 60 minutes
0.005 percentage	Over 60 minutes

3.4.6 Controls. When installed in accordance with applicable drawings, all electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation and under all vehicle operating controls.

3.4.7 Adjustment mechanisms. When installed and adjusted in accordance with applicable drawings, all adjustment mechanisms shall function properly and maintain adjustment settings during all vehicle operating conditions.

3.4.7.1 Driver hatch and driver escape hatch. The force required to breakaway and slide the driver hatch cover across the opening shall not exceed 30 pounds force at the handle. The force required to lock the drive hatch cover in the closed position shall not exceed 75 pounds at the end of the locking level. The driver escape hatch shall seal out water and dirt. The force required to operate the handle to release the driver escape hatch shall be 30 pounds pull at a point 1/2 inch from the end of the handle. Hatch shall fall away without effort when released.

3.4.7.2 Battery access door. The latch that holds door closed shall be free from paint and function properly. The force required to open latch shall not exceed 15 pounds sliding pressure.

3.4.8 Seating. All seat sliding parts and operating contact surfaces shall be free from paint.

3.4.8.1 Driver's seat. With driver in seated position, the force required to actuate the fore and aft seat adjustment lever shall not exceed 12 pounds at the end of the lever. With the driver seat vacated, the force required to actuate the seat vertical adjustment lever shall not exceed 14 pounds at the end of the lever. The force required to trip the seat dumping lever shall not exceed 14 pounds at the end of the lever. The driver seat shall move forward, backward, and vertically when the appropriate levels are actuated. In the lowest position, restriction of the fore and aft movement of the seat is permissable.

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3.4.8.2 Gunner seat. With no load the force required to pull seat adjusting pin shall not exceed 15 pounds. The spring shall raise the seat not less 7 inches without personnel seated.

3.4.8.3 Commander seat.

3.4.8.3.1 Platform seat. The seat shall be capable of sliding upward a minimum of 12 inches with the spring force provided with no load on seat and platform assembly. The no load force required to lift pin for seat adjustment shall not exceed 25 pounds measured at end of lever. The platform latch shall lock securely in "up" position. The backrest shall be retained in a vertical position when not used as a platform. The seat shall be retained in a vertical position when not being used

3.4.8.3.2 Upper swing seat. The force required to position the seat pin for seat release shall not exceed 15 pounds measured at the end of the lever. The force required to rotate the swing seat shall not exceed 10 pounds when applied to the extreme edge of seat.

3.4.8.4 Loader seat. When seat is properly installed and not in use, the lifting spring shall be capable of moving and retaining the seat in a vertical position.

3.4.8.5 Loaders and commanders hatches. With the hatch closed, the force required to unlock the hatch shall not exceed 20 pounds. With the hatch unlocked, the force required to push hatch to its extreme open-latched position shall not exceed 55 pounds measured at the hatch closed lock handle. The force required to unlock hatch from an open latch position and rotate to the balanced position shall not exceed 60 pounds at the end of the hatch opened lock handle. To lock the hatch in the closed position with the hatch closed lock handle, the downward force shall not exceed 50 pounds and the rotational force shall not exceed 20 pounds at the end of the handle.

3.4.9 Electrical system. The vehicle electrical system shall consist of the hull electrical subsystem and the turret and cupola electrical subsystems.

3.4.9.1 Lights. The internal and external lights shall operate as specified on applicable drawings through all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.4.9.2 Interior lighting. Each crew position except commander cupola, shall contain a domelight to provide general illumination to assist with tasks at night or during closed-hatch operation. The domelight shall contain provisions for adjusting light intensity and red filter for night vision adaptation.

3.4.9.3 Hull-to-turret slipring. The hull-to-turret slipring shall provide power and signal circuits capable of maintaining positive electrical contact between the hull and turret electrical subsystems under all vehicle environments. A sealed air passage in the slipring shall be capable

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of carrying low-pressure air from the gas particulate subsystem. Sealed hydraulic fluid passage in the slipring shall be capable of carrying supply pressure, return pressure, and drain pressure for the vehicle hydraulic control system.

3.4.9.4 Cupola slipring. The slipring assembly shall provide uninterrupted power continuity of seven electrical circuits between the turret and cupola. Operation of the slipring shall not introduce static of any kind into the communication system.

3.4.9.5 Auxiliary outlet. Two outlets (24 direct current volts [vdc] nominal), one each in the turret and hull shall be provided that will mate with the OVE trouble light assemblies. Each auxiliary outlet circuit shall be protected with 15 ampere automatic reset circuit breaker.

3.4.9.6 Searchlight system.

3.4.9.6.1 Vehicle and searchlight interface. Vehicle and searchlight requirements without searchlight installed are as follows:

a. The searchlight power circuits shall be capable of providing vehicle power for operating and controlling the searchlight.
b. The searchlight circuit shall be protected with a 60 ampere automatic reset circuit breaker.
c. Mounting provisions shall be provided for searchlight and control box installation.

3.4.9.6.2 Searchlight. With the vehicle in a level position and the main gun at 5 mils elevation and balanced, the installed searchlight shall be capable of the following:

a. Be operable from either the remote control box or the main control box.
b. Have ± 89 mils (± 5°) of adjustment in the elevation and traverse screws.

3.4.9.7 Generator voltage. With the engine operating at a minimum of 1000 revolutions per minute (rpm), the vehicle generator shall deliver between 25.8 volts and 30.2 volts at the battery under all conditions of vehicle operation with the generator fording switch "on". The battery voltage shall be not less than 18.5 volts when the hydraulic power pack is energized.

3.4.9.8 Generator blower motor. The generator blower motor shall be capable of maintaining constant flow of air through generator with the generator fording switch “on”.

3.4.9.9 Passive driver's night viewer power circuit. The voltage output at the passive driver's night viewer connector shall be within 0.2 volts of the positive battery potential when operating into a 1 ampere load.

3.4.9.10 Power plant electrical. The engine electrical installation shall contain the sending units necessary to provide engine data in the form of electrical signals to the driver instrument panel as required and shall include an engine electrical disconnect to facilitate engine

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removal and replacement, and shall also contain an electrical starter motor capable of starting the engine under all required conditions and air cooled generator.

3.4.9.11 Air cleaner blower motors. Each air cleaner blower motor shall be capable of providing a flow of air as specified through each blower outlet. The air cleaner blower motor shall be wired so they do not operate unless the engine is running.

3.4.9.12 Engine manifold heater. The vehicle wiring for the manifold heater shall provide a switch-actuated vehicle battery circuit to the engine. Operation of the push button switch on the purge pump handle with starter switch depressed, shall provide vehicular battery voltage to pin "A" of the multiple connector at the engine electrical quick disconnect.

3.4.9.13 Personnel heater. The heater shall be capable of starting with the switch in either high or low position. The blower motor shall operate at reduced speed when the heater is turned on. The burner shall ignite within 4 minutes of heater turn-on. Subsequent to ignition, the heater shall be capable of generating high and low heat and the blower motor shall run at full speed. Turning the heater "off" shall cause both the burner to extinguish and the blower motor to shut off within 3.5 minutes.

3.4.9.14 Engine smoke generator. The vehicle shall produce smoke in the left and right exhaust pipes when the engine is running and the smoke generator switch is in the "on" position.

3.4.9.15 Communication system. Requirements for a communications system are as follows:

a. An AN/VIC-1 (V) intercom set shall be installed and shall be functional for crew communications.
b. Provisions shall be available for the installation of an AN/VRC radio that shall operate through the intercom set.

3.4.9.16 Main gun firing circuits. The main gun firing circuits shall be capable of providing the energy necessary (1.25 ampere for 100 milliseconds) to fire main round. The firing circuit shall also provide the following:

a. The probe shall be grounded at all times except during application of the firing voltage.
b. The firing control circuits shall include the switching feature which prevents subsequently loaded rounds from detonating until the firing switch has been released after the previous round.
c. Interconnections shall be provided to allow firing circuit activation from the alternate emergency switch on the manual elevation pump handle and provide emergency mode firing per 3.4.9.18.

3.4.9.17 Machine gun firing circuits. The machine gun firing circuits shall be capable of providing the energy necessary (13 amperes) to operate respective machine gun solenoids.

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3.4.9.18 Main gun and machine gun circuit controls. The firing circuit controls shall be arranged and interlocked to allow the commander to fire the main gun, coaxial machine gun, and the cupola machine gun and to allow the gunner to fire the main gun and the coaxial machine gun. The main gun shall be provided with production against accidental firing by a loader's safety switch which interrupts the commander and gunner firing circuits automatically during gun recoil. The switch shall thereafter remain in the safe (open circuit) position until manually returned to the fire (closed circuit) position by the loader. The blasting machine shall provide an alternate source of electrical power for emergency firing of the main gun through the loader safety switch.

3.4.9.19 Smoke grenade discharger circuits.

3.4.9.19.1 Arming. The power-on (ready) lamp in the grenade power box shall illuminate when the master battery switch, turret power switch, and the grenade system power switch are all on.

3.4.9.19.2 Circuit voltage. Voltage at the smoke grenade discharge firing pins shown in figure 6 shall be as indicated in table I when the power-on (ready) lamp in the grenade power box is illuminated and the appropriate firing pushbutton is actuated.

TABLE I – Grenade discharger circuit voltage.

Firing pushbutton	Left discharger firing pins	Right discharger firing pins	Voltage
Right	1, 2, 5	3, 4, 6	Vehicle (Ref. 3.4.9.7)
Right	3, 4, 6	1, 2, 5	0 ± 0.5 VDC
Left	3, 4, 6	1, 2, 5	Vehicle (Ref. 3.4.9.7)
Left	1, 2, 5	3, 4, 6	0 ± 0.5 VDC

3.4.10 Vision devices and receptacles. Provisions shall be made to assure that receptacles incorporated in the vehicle will except applicable vision devices without binding or interference. The gunner and commander periscope receptacles shall accept the periscopes, and when installed, removed, and reinstalled the line of sight shall not deviate more than 0.2 mil from the line of sight of initial installation setting. When installed, the periscope receptacles, vision devices, and vision blocks shall prevent the entrance of water into vehicle.

3.4.11 Welding repairs. Welding repairs of any type or class shall be made only when specifically authorized by the acquisitioning activity.

3.5 Break-in run.

3.5.1 Preparation. Prior to break-in run, the vehicle shall be completely assembled and adjustments completed. Vehicle, including all components requiring lubrication shall be lubricated and serviced with fuel as directed (see 6.4). Lubricants and fluids shall be as specified in Production Lubrication Drawing D10940135 detail of EPL 8736334. Engine oil conforming

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to the applicable seasonal grade of MIL-L-2104 can be used. Specific applications by grades or types for seasonal temperature ranges shall be as specified (see 6.4). Fuel and water separator and primary fuel filter shall be bled of air by loosening each respective bleed plug and operating the in-tank fuel pump. After air is bled, retighten the bleed plugs.

3.5.2 Oil pressure and temperature. Prior to start of the break-in run, the engine shall be operated at 800 to 1200 revolutions per minute (rpm) until lubricating oil is at operating pressure and temperature. If during break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated as above before continuing break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60 percent grades, and 30 percent side slope.

3.5.3 Operation and distance. Prior to tests, each vehicle shall be given a break-in run for the distance specified in table II on smooth, level, hard-surfaced roads. The vehicle shall be operated over the undulating section of road (see figure 5) during "a" and "b" divisions of the break-in run in both directions.

TABLE II. Speeds and distances for break-in run.

Division of run	Speed miles per hour (mph)	Distance miles	Test conditions
a.....	0 to 10	20	Track (road)
b.....	10 to 20	20	Track (road)
c.....	21 to max	10	Track (road)

3.5.4 Reverse operation. After each division of the break-in run, vehicles shall be stopped, engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet at a speeds between 2 and 7 mph.

3.5.5 Condition after run. After completion of the break-in, the engine shall idle between 700 and 750 rpm. Prior to vehicle being submitted to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct. There shall be no defects or maladjustments that may cause faulty vehicle operation.

3.6 Performance. Complete vehicle, combat loaded or with a simulated load of equal weight (see 3.4.2) and serviced, shall perform as specified herein. Ammunition may be simulated in size and weight. Vehicle serviced (see 3.5.1) and equipped for existing climatic conditions shall operate as specified without special equipment.

3.6.1 Environmental. Vehicle shall operate in ambient air temperature of 115°F to minus 25°F, and with special equipment installed, the vehicle shall operate at temperatures between minus 25°F and minus 65°F. Complete vehicle shall withstand storage between minus 65°F and 155°F without deterioration that may cause failure of any component part of the vehicle.

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3.6.2 Power package and train. When power package is installed in accordance with applicable drawings, it shall function throughout all gear and speed ranges without loss of lubricants. The power train and associated controls shall be capable of operation throughout all speed and steering ranges without binding of linkages, loss of lubricants, grabbing, and chattering or slippage when controls are applied to steer, stop, and hold vehicle.

3.6.3 Speeds.

3.6.3.1 Level road speeds. Vehicle shall operate at sustained speeds of 30 mph and at 2.5 mph on level ground without damage to power plant and power train. When traveling between 25 and 30 mph in a straight line, the vehicle drift shall not exceed 3 feet in one 100 feet.

3.6.3.2 Grade speeds. Vehicle shall operate at a sustained speed of 10 mph while ascending a 10 percent grade, and at 18 mph while ascending a 3 percent grade.

3.6.4 Acceleration. Vehicle shall accelerate from a standing start on level ground through a distance of 200 feet in not more than 13 seconds.

3.6.5 Engine starting on grades and slopes. With the vehicle standing on a 60 percent grade for not less than 2 minutes, and engine operating under no load between 700 and 750 rpm, the engine shall be stopped for not less than 2 minutes. The engine shall restart in not more than 1 minute when on up and down grade. Similar engine stopping and starting operation shall be performed on 30 percent right and left side slopes.

3.6.6 Braking and drift.

3.6.6.1 Stopping. Vehicle traveling at 20 mph shall stop within 60 feet from point of brake application; drift not to exceed 4 feet when stopping.

3.6.6.2 Holding. With vehicle combat loaded or with a simulated equal weight and standing on a 60 percent grade with service brakes applied, the vehicle shall be held stationary when on up and down grade. With parking brake engaged and all other holding devices inoperative, the vehicle shall be held stationary when on up and down grades.

3.6.7 Turning. Vehicle shall pivot 360° to the right or left, in neutral steer range, within a circle 35 feet in diameter.

3.6.8 Fording.

3.6.8.1 Shallow water. Vehicle shall ford a level, hard-bottom body of water 48 inches in depth including wave without special equipment. With vehicle standing in water 48 inches in depth for 30 minutes, the accumulation of water shall not be more than 1-1/2-inch on the crew compartment hull floor measured in the center of the "v" and there shall be no water accumulation in the right or left brake housings (see 6.6).

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3.6.8.2 Deep water. Vehicle shall ford a level, hard bottom, water 8 feet in depth when a fording kit is installed. With the vehicle standing in water 8 feet in depth for 30 minutes and the bilge pump operating as required, the accumulation of water shall not be more than 3 inches on the crew compartment hull floor in the center of the "v" and there shall be no water accumulation in the right or left brake housings (see 6.6)

3.6.8.3 Front and rear drain valves. After unlocking level, the force required to operate the front drain valve level shall not exceed 17 pounds at the end of the hand grip to actuate the valve a minimum of 3/8 inch travel. After unlocking lever, the force required to operate the rear drain valve level shall not exceed 25 pounds at a point just below the knob to actuate the valve through a minimum of 11/16 inch travel.

3.6.8.4 Inflatable seal and pump. The pump shall be capable of inflating the seal to 25 psi. The force required to operate the pump plunger shall not exceed 50 pounds.

3.6.8.5 Engine starting in shallow water. With vehicle standing for 30 minutes in water 48 inches in depth, the engine operating at 1,000 rpm for 15 minutes, then stopped for 15 minutes, the engine shall restart in not more than 3 minutes. All accessories shall function satisfactorily during and after fording operations. After fording operations, the water contamination content of the transmission, engines, final drives, and suspension system lubricants shall not be more than 2 percent by volume.

3.6.9 Trench crossing. Vehicle shall cross trenches 36 inches in depth and 99 inches in width without stalling or damage to suspension, boom, or fenders.

3.6.10 Vertical obstacles. Vehicle shall cross over vertical obstacles 30 inches in height while moving forward without stalling or damage to suspension and hull floor.

3.6.11 Climbing. Vehicle shall ascend longitudinal grades of 60 percent in forward and reverse gear without stalling or damage to power plant and power train. Vehicle shall operate on right and left side slopes of 30 percent.

3.6.12 Turret and gun control system and gun fire control system.

3.6.12.1 System performance. Unless otherwise specified, the turret and gun control system and gun fire control system shall operate and meet all requirements specified herein for the following operating conditions:

a. The turret shall be level within 1 degree.
b. The vehicle electrical system shall supply 26 vdc minimum to the turret and gun control system with engine operating and the generating system on.
c. The main armament shall be balanced as specified in 3.6.12.2.
d. The gun shall operate throughout the full range of depression an elevation.

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e. The turret shall traverse in either direction throughout 360 degrees.
f. The boom assembly shall be in the stowed position and locked to the turret.

NOTE: The systems shall be capable of operating at 18 to 30 vdc.

3.6.12.2 Main armament balance. With the co-axial machine gun, spent brass and link ejection bag, deflector, convertor for auxiliary firing system, telescope, nylon ballistic shield, mantlet cover, OVE or equal compensating weights installed with a dummy projectile weighing 65 pounds in the main armament, and with the gun at the horizontal position within one-half degree, the main armament shall be balanced by the gun equilibrator pressure to obtain the manual elevation hand crank torque reading within 3 1/2 inch-pounds to elevate and depress the gun one degree from the horizontal position.

3.6.12.3 Nylon ballistic shield. Gun mount conforming to MIL-M-45212 with co-axial machine gun mount, M73 machine gun, spent brass and link ejection bag, sighting system, and nylon ballistic shield installed, shall elevate to a minimum of 20 degrees and a minimum of 10 degrees depression. No binding shall exist between the machine gun, sighting system, and the nylon ballistic shield, to cause the co-axial machine gun to deflect by more than 1 mil and the telescope to deflect more than 0.1 mil.

3.6.13 Turret and gun traverse and elevation control system.

3.6.13.1 Manual traverse effort. The mean torque applied to the manual traverse hand crank, at a uniform rate to maintain turret movement shall not exceed 32 inch-pounds.

3.6.13.2 Manual elevation effort. The mean torque required to rotate the manual elevation hand crank at a uniform rate to maintain gun movement shall not exceed 46 inch-pounds. Individual readings shall not exceed 55 inch-pounds.

3.6.13.3 Manual elevation and depression response rate. The gun at the horizontal position within plus or minus one degree, shall move at a minimum rate of 10 mils per revolution of the hand crank in elevation and depressions. The hand crank shall be turned at a rate not less than 10 rpm and not more than 20 rpm to measure the response rate.

3.6.13.4 System backlash. With the traverse and elevation power switch on or off, the control system backlash shall not exceed one mil in traverse and one mil in elevation.

3.6.13.5 Power and manual control. The manual traverse and power elevation controls shall be capable of simultaneous operation. The same requirement applies to the power traverse and the manual elevation controls. There shall be no movement of the gun or turret caused by turning the elevation and traverse power switch on or off. The gun firing switches shall function individually with elevation and traverse power switch on or off and with the main gun or coaxial machine gun selector switches turned on individually or collectively.

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3.6.13.6 Override control. The commander control shall take instantaneous power control of the turret and gun and the firing circuits from the gunner control when the override switch in the commander control handle is actuated. Override of the system control by the commander control shall be obtainable at the neutral position regardless of the handle position of the gunner control. With the commander control and gunner control in the neutral position, there shall be no movement of the gun or turret caused by depressing or releasing the override switch in the commander control handle. With the gunner control in neutral position, gunner control shall regain system control instantaneously when the commander control handle is released.

3.6.13.7 Control system deadspot. The turret and gun control system deadspot as measured at the gunner and commander power controls shall not exceed 7° handle movement from the neutral center position in any direction to initiate gun and turret movement. The neutral center position shall be defined as that handle position which divides the initial free handle movement (backlash of linkages) in half. The deadspot angles on each side of the neutral center in the traverse axis shall be equal within 2° and the deadspot angles in the elevation axis shall be equal within 2° for both the gunner and commander power control.

3.6.13.8 Gun elevation speeds. The gun shall be controllable in elevation and depression at all speeds between 0.5 mil per second and 71 mils per second (4° per second). These requirements are applicable to both the gunner and commander controls.

3.6.13.9 Turret traversing speeds. The turret shall be controllable in traverse at all speeds between 0.5 mil per second and 170 mils per second. These requirements apply to both gunner and commander power controls.

3.6.13.10 Main gun stability tests. With the vehicle level or on slopes up to and including 15° turret attitude, and turret power activated or deactivated, the gun shall not move more than 0.5 mil in azimuth or elevation during a 1 hour period.

3.6.13.11 Elevation, depression and traverse limits. The angular gun movement and vehicle clearance shall be in accordance with the following requirements:

a. At a minimum elevation of 20°, the main gun shall be capable of being traversed for 360° by power and manual controls with the elevation and traverse power switch on. The traverse shall be accomplished without vehicle interference.

b. At a minimum depression of 10°, the main gun shall be capable of being traversed within the zone of 90° each side of the vehicle front centerline by power and manual controls with elevation and traverse power switch on. The traverse shall be accomplished without vehicle interference.

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c. At a minimum depression of 5°, the main gun shall be capable of being traversed within the zone from 90° each side of the vehicle rear centerline to 20° each side of vehicle rear centerline by manual control with elevation and traverse power switch off. The traverse shall be accomplished without vehicle interference.

d. At full depression the main gun shall be traversed at maximum traverse speed within the zone of 90° each side of the vehicle front centerline with elevation and traverse power switch on. The main gun shall automatically elevate outside of this zone so that the traverse is accomplished without vehicle interference.

e. At an elevation of 0° the main gun shall be capable of being traversed within the zone of 90° each side of the vehicle rear centerline with elevation and traverse power switch on. The traverse shall be accomplished without vehicle interference.

3.6.13.12 Gun laying (stationary target, vehicle stationary). From a layoff position tabulated in table III, the average time required for a total of four, eight, or 12 trials to position the gunsight reticle within the borders of a 0.25 mil square target shall not exceed that tabulated in table III. The time for layoff position number three shall be met for an equal number of trials from each of four different quadrants from target. The time of all other layoff positions shall be met for two directions from targets. These requirements shall be met by gunner and commander power and manual controls as stipulated in table III. The results of any series of trials shall not be used in any other series.

Table III Gun laying on stationary target.

Layoff - mils			Average time - seconds	
Position	Azimuth	Elevation depression	Power controls	Manual controls
1	-	± 10	2	2
2	25	-	3	6
3	25	± 10	5	8
4	100	-	5	
5	400	-	7	

3.6.13.13 Turret and gun control system operation on slope. When the vehicle is canted or pitched on slopes up to and including 15° turret attitude, the gun control system shall be capable of gun laying on a 0.25 mil stationary target as specified in 3.6.13.12, table III, except time limitation, need not be met. This requirement shall apply only to the gunner power control. The gun control system performance on level terrain shall not be affected in any manner as a result of vehicle operation on slopes. Combat load of simulated equipment may be employed if desired.

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3.6.13.14 Travel locks. The turret with turret lock engaged shall be prevented from being traversed in any direction. With the gun external travel lock and the turret lock disengaged and the manual traverse handle in detent, the turret and gun control system shall act as a holding device for the gun and turret when the vehicle is in motion.

a. With the vehicle moving at a rate of 5 mph in low gear on a horizontal hard surface road, and with the elevation and traverse power switch off, the turret slippage shall not exceed 10 mils in traverse when subjecting the vehicle to a full steer 90° turn in either direction and coming to a sudden stop.

b. With the vehicle moving at a rate of 20 mph on a horizontal, hard surface road and with the elevation and traverse power switch on or off, the gun slippage shall not exceed 5 mils in elevation or depression when stopping the vehicle within 60 feet from the point of vehicle brake application.

c. The elevation hand crank shall not show any movement by turret induced reaction when vehicle is in motion as specified in (a) and (b) above, or by power operation of the turret and gun control systems.

3.6.13.15 Hydraulic pressure limit switch. When power traversing the turret at a rate not to exceed 65 mils per second, the hydraulic pressure switch shall activate the hydraulic control powerpack motor at a control system pressure gage reading of 880 to 1010 psi. With gunner and commander power and manual controls in neutral position, the hydraulic pressure switch shall deactivate the hydraulic powerpack motor control system pressure as indicated on the gage between 1150 and 1500 psi.

3.6.13.16 Elevation shut-off valve. With elevation and traverse power switch on, the movement of the main gun shall not exceed 2 mils in the elevation axis when the gunner power control handle is tilted forward and backward to its stop in either direction without actuating the palm switch in the gunner power control handle.

3.6.13.17 Turret traverse brake. With the elevation and traverse power switch on, and with the turret stationary, there shall be no traverse movement of the turret when the gunner power control handle is turned to its stop in a clockwise or counterclockwise traverse direction without actuating the palm switch in the gunner power control handle.

3.6.13.18 Hydraulic fluid.

3.6.13.18.1 Turret and gun control cleanliness. When all the requirements of the turret and gun control system have been met, and with hydraulic fluid discharged from the 2 gallon main accumulator and from the 1/2 gallon equilibrator accumulator, the hydraulic fluid in the control powerpack reservoir and the equilibrator charging manifold reservoir shall contain no more than 0.1 percent water contamination level and shall not exceed that shown in table IV.

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Table IV – Maximum contamination limit (100 ML sample).

Particle size range (Microns)		Number of particles
Turret & gun control	Vehicle hydraulic control	
5-15		128,000
15-25		22,000
25-50		4,050
50-100		720
Over 100		128
	25-50	4,050
	50-100	720
	Over 100	128

3.6.13.18.2 Leakage. Turret and gun control system hydraulic fluid leakage shall be permissible in the form of a weep or seep. There shall be no evidence of fluid leakage in the form of a drop or drip other than that specified herein. There shall be no drip from the deck clearance valve assembly. Fluid leakage (drip) from the control power pack at the override linkage control box shall not exceed one drop in 2 minutes. Fluid loss from the main gun recoil mechanism or replenisher shall not exceed 15 drops in a 3 minute period during and immediately after the completion of firing or exercising of the recoil mechanism. No fluid loss in the form of a drip or drop shall be permitted 2 hours subsequent to firing or exercising the main gun.

3.6.14 Fire control systems. Unless otherwise specified, the fire control system shall operate and meet all requirements specified herein under operating conditions specified in 3.6.12.1.

3.6.14.1 Synchronization of gunner periscope and gun tube bore. The line of sight of the gunner periscope (daylight system only) shall move in synchronization with the centerline of the gun tube bore within plus or minus 0.3 mil in both elevation and deflection in all gun positions from horizontal to 15° elevation. The telescope shall move in synchronization with the centerline of the gun tube bore within plus or minus 0.1 mil in both elevation and deflection at 15° elevation and 5° depression. The gun tube bore centerline at 15° elevation shall be within 0.5 mil of a plane perpendicular to the plane of traverse and coincident with the line of sight of gun when gun is at zero elevation.

3.6.14.2 Elevation backlash. Elevation backlash in the gunner periscope shall not exceed 0.3 mil for any gun position from 10° depression to 20° elevation.

3.6.14.3 Boresight knob travel. The boresight knobs on the gunner periscope and the telescope mount shall be adjustable to a minimum of 5 mils down, 2 mils up, 4 mils left, and 4 mils right as measured from the boresight position.

3.6.14.4 Unity sight knob travel. Rotation of the gunner unity sight knobs in elevation and deflection from stop to stop shall cause the reticle to travel a minimum of 20 mils.

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3.6.14.5 Boresight retentions. The gunner periscope and telescope shall maintain previously established boresight within the values of table V.

Table V - Boresight retention.

Periscope		Telescope	
Elevation	Deflection	Elevation	Deflection
+ 0.25 Mils	+ 0.25 Mils	+ 0.15 Mils	+ 0.15 Mils

3.6.14.6 Night vision. Actuation of the night vision channel switch of the gunner periscope or the commander periscope shall remove the shutter from the optical path and shall provide low light level operation viewing and sighting through the night vision channel. Image brightness shall be maximum with the tube control knob in the extreme clockwise position. The reticle shall not be illuminated with the reticle control knob in the "off" position, and it shall be of maximum brightness with the reticle control knob in the extreme clockwise position.

3.6.15 Cupola performance.

3.6.15.1 Slip ring contact boards. The slip ring contact boards mounted in the turret shall be vertically adjusted so that the bottom edge of the lowest contact spring is in the same horizontal plane as the bottom edge of the lowest contact strip within plus or minus .010 inch.

3.6.15.2 Slip ring circuits. When the completed cupola has been assembled to the turret, it shall include slip rings to provide uninterrupted continuity of communications, power, and ground circuits throughout continuous 360° of cupola traverse. Each slip ring power circuit shall be capable of transferring between turret and cupola a minimum of 15 amperes of constant current. Each slip ring communication circuit shall be capable of transferring a minimum of 10 amperes continuous current.

3.6.15.3 Slip ring noise. The operation of slip ring shall not introduce static of any kind into the communications system.

3.6.15.4 Traverse effort. Azimuth gear box starting torque, for right or left traverse, shall be not more than 30 inch-pounds to start motion from any position within 360° rotation.

3.6.15.5 Elevation effort. With machine gun flexible feed chute, canvas cover and periscope linkage installed, the screwjack handle breakaway torque shall not exceed the following:

15° depression to 60° elevation, 25 in. lb.
60° elevation to 10° depression, 25 in. lb.
10° depression to 15° depression, 35 in. lb.

3.6.15.6 Elevation limits. The cradle with machine gun sighting system and cradle cover installed shall have a minimum of 60° elevation and a minimum of 15° depression.

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3.6.16 Indirect fire control.

3.6.16.1 Elevation quadrant. At zero gun elevation and with the bubble centered in the level vial, the knob scale and index scale shall be adjusted to read zero. The reading on the quadrant shall correspond to the position of the gun within plus or minus 1 mil at all angles of elevation and depression.

3.6.16.2 Azimuth indicator backlash. The backlash of the azimuth indicator shall not exceed 0.5 mil in any position of the turret.

3.6.16.3 Leveling devices. The level on the ballistic drive shall be level within 15 minutes (three graduations) when the gun is at the boresight position.

3.6.17 Cupola weapon sighting system.

3.6.17.1 Synchronization. The periscope line of sight shall be synchronized with the cupola machine gun so that the line of sight of the periscope maintains a given azimuth and elevation relationship (established at horizontal gun position) with the centerline of the machine gun bore within 1.8 mils at all gun positions between 10° depression and 30° elevation, and within 3 mils at all gun positions between 30°, and 50° elevation.

3.6.17.2 Backlash. The elevation backlash of the cupola machine gun sighting system shall be not more than 0.6 mil at any position exclusive of machine gun free barrel movement.

3.6.17.3 Boresight knob travel. The boresight knobs on the periscope shall be adjustable to a minimum of 5 mils down, 2 mils up, 4 mils left, and 4 mils right as measured from the boresight position.

3.6.17.4 Boresight retention. The line of sight of commander periscope shall maintain previously established boresight within 1.5 mils.

3.6.17.5 Interlock assembly location. The interlock assembly shall be located for azimuth coincidence of the machine gun line of sight and the main armament line of sight at 600 meters within 2 mils.

3.6.18 Support equipment hydraulic control system.

3.6.18.1 Vehicle hydraulic control system. The vehicle hydraulic control system shall control the operation of the bulldozer, winch, and boom, and shall meet all requirments specified herein at an engine idle speed of 1100 rpm.

3.6.18.2 Boom. The boom shall be hydraulically raised from the travel locks to a vertical position in not more than 45 seconds. The winch shall return the boom to within 6 inches of stowed position.

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3.6.18.3 Winch. The winch shall be capable of reeling the cable on the fourth layer at a rate of not less than 30 feet per minute in low gear with the engine at 1800 rpm. The winch shall produce a line pull of not less than 17,500 pounds using a single part line in low gear, fourth layer of cable on the drum. The hydraulic relief valve shall prevent the exertion of a line pull of 20,000 pounds or more under the same conditions.

3.6.18.4 Winch brakes. The brakes of the winch shall be a type having automatic mechanical application and hydraulic release. The brake shall automatically apply and hold the load and prevent winch drum turning after hydraulic power has been removed. With the weight suspended off the ground and winch control in neutral, there shall be no downward movement of weight.

3.6.18.5 Bulldozer. The bulldozer moldboard movement shall be controlled by a four position control valve. Positions shall be: raise, hold, lower, and float. The "raise" position shall move the moldboard upward at a minimum rate of 2.5 inches per second at engine speed of not more than 1200 rpm from ground level to full raised position. The "hold" position shall maintain the moldboard in approximate mid-position between ground level and carrying hook height for 1 minute without the moldboard drifting downward more than 1/4-inch. The "lower" position shall lower the moldboard from the maximum raised position to 6 inches below ground level. In the maximum "lower" position, the bulldozer shall be capable of lifting the front of the vehicle and hold in this position for 1 minute. The "float" position shall allow the moldboard to drift downward from raised position to ground level.

3.6.18.6 Vehicle hydraulic control system cleanliness. When all the requirements of the vehicle hydraulic control system have been met, the hydraulic fluid in the reservoir shall contain no more than 0.1 percent water and the contamination level shall not be greater than that shown in table IV.

3.6.18.7 Hydraulic system fluid leakage. Fluid leakage of the vehicle hydraulic control system components, excluding lines and fittings shall be permissible in the form of a weep, seep, or droplet. There shall be no evidence of fluid leakage in the form of a drip from the vehicle hydraulic control system.

3.7 Painting, marking and data plates.

3.7.1 Painting. Unless otherwise specified, the exterior and interior of vehicle and all components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and exterior non-skid paint shall be forest green camouflage lusterless enamel, alkyd, per MIL-E-52798. The vehicle interior paint shall match color chip 17875 of Federal Standard No. 595. All sliding parts and operating contact surfaces shall be free of paint.

3.7.2 Marking. Marking and marking paint shall conform to Federal Specification TT-E-527, lusterless black, color chip 37038 of Federal Standard 595 as specified on drawing 8736334.

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3.7.3 Name, shipping, and service data plates. All plates shall be in accordance with the applicable drawings and MIL-P-514.

3.8 Fire extinguisher. The fixed fire extinguisher system installed in accordance with applicable drawings shall be operational internally and externally. Actuating the internal controls shall simultaneously stop the supply of fuel to the engine. The maximum effort required to discharge the system internally or externally, shall not be more than 55 pounds. A time delay of not more than 6 seconds but less than 11 seconds shall occur on the first shot between actuating the release mechanism and entrance of CO_2 into the engine compartment.

3.9 Gas particulate filter system.

3.9.1 Air flow. A minimum air flow of 3.5 cubic feet per minute (CFM) is required at each crew position outlet when the other crew position hoses are connected to their respective outlets.

3.9.2 Air heater. An air heater will be supplied at each crew position. When the heater switch is turned on, the heater light must illuminate. The heater must warm the air a minimum of 10°F over ambient air within a 10 minute period at an input voltage of 24-30 volts. Thermostats of each heater must function to maintain temperature settings of their control knobs.

3.10 Electromagnetic compatibility (EMC) characteristics. The vehicle shall meet the requirements of MIL-STD-461A. Notice 4 as follows:

3.10.1 Radiated emissions. Test RE05, of MIL-STD-462, Notice 3.

3.10.2 Radiated susceptibility. Test RS03, of MIL-STD-462, Notice 3.

3.10.3 Conducted emissions. Test CE07, of MIL-STD-462, Notice 3.

3.10.4 Electromagnetic compatibility. Operation of any subsystem or equipment in its design mode of operation, or while switching modes of operation shall not result in undesirable response of any other subsystem or equipment beyond those criteria determined by vehicle test.

NOTE: Radiated emissions, radiated susceptibility, conducted emissions and electromagnetic compatibility shall be determined by analysis of data obtained from electromagnetic interference (EMI) and EMC test of production vehicles per 4.5.3.12, MIL-STD-461A, and MIL-STD-462.

3.11 Vibration and noise level.

3.11.1 Equipment operation. Equipment operation shall not exceed the maximum acceptable noise level specified in table VI.

3.11.2 Operator's protection. Noise hazard caution signs shall be posted in all crew stations. The signs shall be clearly visible to all personnel.

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Table VI. Maximum acceptable noise levels for Army Materiel Command equipment (continuous noise)

Octave band Center frequency (Hz)	Maximum acceptable noise level (db ref. 0.0002 microbrar)
63	121
125	111
250	103
500	102
1000	100
2000	100
4000	100
8000	100

3.12 Workmanship. Workmanship shall be of a quality to assure that vehicle and components are free from defects resulting from improper manufacturing or assembly practices and shall conform to requirements herein specified.

3.13 Recycled, virgin and reclaimed materials. The use of recycled materials which meet the requirements of the applicable item or part without jeopardizing its intended use shall be encouraged.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. Unless otherwise specified by the acquisition activity, the contractor shall provide and maintain an effective Inspection and quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted by the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced herein.

4.1.2 Government verification. All quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written

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inspection plan are being properly applied and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures that might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Parts and components. Parts, components, assemblies and installations shall be inspected in accordance with the drawings listed on EPL 8736334 and Master list of Supplementary Quality Assurance Provisions 8736334 (see 3.4).

4.1.4 Qualified products and materials. The contractor's inspection records shall be checked to determine contractor conformance to 3.3.1 and 3.3.2.

4.1.5 Classification of inspections.

a. Preproduction and first production (see 4.2).
b. Quality conformance inspection (see 4.5).

4.2 Preproduction and first production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production inspections.

4.2.2.1 Special process inspection. A special process inspection shall be conducted by the Government representatives during fabrication of the first production vehicle (see 3.2) to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems, and plan of quality control shall be conducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.2.1.2 Hull and turret inspection. The hull and turret shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.2.2 Complete vehicle inspection. To determine conformance to 3.2, contractor shall perform acceptance and control inspections listed in table VIII on the first completed vehicle. Subsequent to contractor's inspection the Government will inspect the vehicle at the contractor's plant.

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4.2.2.2.1 Corrections. Corrections shall be made by contractor on the vehicle prior to acceptance of the vehicle by Government.

4.2.3 Production vehicle test. After completion of contractor's inspection on the first completed vehicle, a second vehicle, or vehicles, shall be selected from the first ten vehicles produced, and tested to establish conformance to 3.2.1 and 3.2.1.1. The vehicle(s) shall be subjected to examinations and tests as specified in 4.4a, b, and c. Subsequent to these tests, the selected vehicle(s) shall be subjected to the 4,000 mile test specified in table VII. Tests shall be performed by the Government at approved Government site with 90 day maximum requirement. Delays, due to poor quality of workmanship or materials, or failure of the contractor to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract performance dates, delivery schedule, or contract price.

TABLE VII. 4,000 Mile combat loaded test
(Government proving grounds)

Course	Mileage and speeds
Hard surface roads	1,000 miles at varying speeds up to maximum
Gravel and dirt roads	1,000 miles at varying speeds up to maximum
Level cross country	1,000 miles at varying speeds up to maximum
Hilly cross country	1,000 miles at varying speeds up to maximum

4.2.3.1 Test failure. Failure of the vehicle as the result of any deficiency of workmanship or materials during or as a result of the 4,000 mile test and hydraulic system test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 4,000 mile and hydraulic system test shall be prima facie evidence that all vehicles already accepted prior to completion of the 4,000 mile and hydraulic system test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 In-process inspection. In-process inspection shall be performed to determine conformance to applicable requirements prior to submitting a vehicle for acceptance inspection. The in-process inspections to be performed are as follows:

a. Parts and components (see 4.1.3 and 4.5.2).
b. Hull and turret inspection (see 4.2.2.1.2).
c. Break-in run (see 3.5 and 4.4).
d. Welding repairs (see 3.4.11).

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e. Fuel tank hydraulic oil reservoir, and line cleanliness (see 3.4.4.1, 3.4.4.2 and 4.5.1.5).
f. Heater fuel feed (see 3.4.4.3 and 4.5.1.6).
g. Battery door check (see 3.4.7.2 and 4.5.1.11).
h. Generator voltage check (see 3.4.9.7 and 4.5.1.14).
i. Generator blower motor check (see 3.4.9.8 and 4.5.1.15).
j. Main armament balance check (see 3.6.12.2 and 4.5.1.40.1).

4.4 Break-in run, track. To determine conformance to 3.5.1, 3.5.2, and 3.5.3 each vehicle shall be lubricated and serviced prior to the break-in run (see 6.4). The run shall be on smooth, level, hard-surface roads, and for the distances specified in table II. The vehicle shall be operated in reverse after each division as specified in 3.5.4. The undulating portion of the test shall be accomplished twice in each direction at speeds between five and 10 mph during division "a".

4.4.1 Condition after run. The vehicle shall be inspected to determine conformance to 3.5.5. End play check shall be accomplished on all suspension wheel bearings and may be readjusted if required.

4.5 Quality conformance inspection.

a. Acceptance inspections 4.5.1 through 4.5.1.43.5.
b. Control inspections 4.5.2 through 4.5.2.12.
c. Comparison inspections (CT) 4.5.3 through 4.5.3.14.

4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.6 after break-in run (see 3.5), each vehicle shall be operated for a distance of 8 miles or more and shall be examined as specified in 4.4.2 and subjected to acceptance tests (see 4.5). These acceptance test shall include all acceptance tests specified at place of manufacture in table IX. Vehicle shall be completely assembled serviced with fuel, lubricants, and fluid as specified, but need not be combat loaded, except for holding as specified in 4.5.1.40. After tests, the vehicle shall be examined for evidence of fuel or lubricant leakage or other deficiencies.

4.5.1.1.1 Test failure. If a vehicle fails to pass any test specified herein, the Government inspector shall withhold acceptance of subsequent lots until evidence has been provided by the contractor that the corrective action has been taken.

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TABLE VIII. Location for tests.

Title	Paragraph No.	Place of manufacture	Government proving ground
Acceptance tests	4.5.1.1	X	X
Sealing check	4.5.1.2.1	X	X
Hatch seal test	4.5.1.3	X	X
Fuel system slope check	4.5.1.6	X	X
Fuel line check	4.5.1.5.1	X	X
Hydraulic system check	4.5.1.5.2	X	X
Heater fuel feed check	4.5.1.6	X	X
Control checks	4.5.1.7	X	X
Adjustment mechanism check	4.5.1.8	X	X
Drivers hatch and escape hatch check	4.5.1.9	X	X
Loaders and commanders hatch check	4.5.1.10	X	X
Seating check	4.5.1.12	X	X
Light check	4.5.1.13	X	X
Passive driver's night viewer power circuit check	4.5.1.16	X	X
Slip ring and external communication circuit check	4.5.1.17	X	X
Auxiliary outlet check	4.5.1.18	X	X
Searchlight control system check	4.5.1.19	X	X
Power plant electrical check	4.5.1.20	X	X
Air cleaner blower check	4.5.1.21	X	X
Engine manifold heater check	4.5.1.22	X	X
Personnel heater	4.5.1.23	X	X
Engine smoke generator test	4.5.1.24	X	X
Communication system check	4.5.1.25	X	X
Main gun firing circuits check	4.5.1.26	X	X
Machine gun firing circuits check	4.5.1.27	X	X
Main gun and machine gun circuit controls check	4.5.1.28	X	X
Smoke grenade discharger circuit check	4.5.1.29 thru 4.5.1.29.3	X	X
Power package and train test	4.5.1.30	X	X
Level road speed test	4.5.1.31	X	X
Acceleration test	4.5.1.32	X	X
Engine starting on grades and slopes test	4.5.1.33	X	X
Stopping test	4.5.1.34	X	X
Holding	4.5.1.35	X	X
Turning test	4.5.1.36	X	X
Fording test	4.5.1.37	X	X
Drain valve checks	4.5.1.38	X	X
Inflatable seal and pump check	4.5.1.39	X	X

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TABLE VIII. Location for tests – Continued.

Title	Paragraph No.	Place of manufacture	Government proving ground
Turret and gun control system check	4.5.1.40	X	X
Nylon ballistic shield check	4.5.1.40.2	X	X
Manual traverse effort test	4.5.1.40.3	X	X
Manual elevation effort test	4.5.1.40.4	X	X
Manual elevation and depression response rate test	4.5.1.40.5	X	X
System backlash test	4.5.1.40.6	X	X
Simultaneous use of power and manual control test	4.5.1.40.7	X	X
Override control test	4.5.1.40.8	X	X
Control system deadspot test	4.5.1.40.9	X	X
Gun elevation speed test	4.5.1.40.10	X	X
Turret traversing speed test	4.5.1.40.11	X	X
Main gun stability test	4.5.1.40.12	X	X
Elevation and depression limit tests	4.5.1.40.13	X	X
Gun laying on stationary target test	4.5.1.40.14	X	X
Operation on slope test	4.5.1.40.15	X	X
Turret locking test	4.5.1.40.16	X	X
Azimuth indicator test	4.5.1.40.17	X	X
Turret control system fluid leakage test	4.5.1.40.18	X	X
Hydraulic pressure limit test	4.5.1.40.19	X	X
Gun fire control system test	4.5.1.41	X	X
Synchronization test	4.5.1.41.1 thru 4.5.1.41.1.2	X	X
Sighting system backlash	4.5.1.41.2	X	X
Boresight knob travel test	4.5.1.41.3	X	X
Leveling devices test	4.5.1.41.4	X	X
Elevation quadrant M13A3 test	4.5.1.41.5	X	X
Cupola slip ring contact boards test	4.5.1.42.1	X	X
Cupola slip ring circuits test	4.5.1.42.2	X	X
Cupola noise at slip ring test	4.5.1.42.3	X	X
Cupola azimuth gear box hand-wheel effort test	4.5.1.42.4	X	X
Cupola interlock assembly location test	4.5.1.42.5	X	X
Cupola machine gun sighting system tests	4.5.1.42.6	X	X
Cupola machine gun synchronization tests	4.5.1.42.6.1	X	X
Backlash test	4.5.1.42.6.2	X	X
Boresight knob travel	4.5.1.42.6.3	X	X

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TABLE VIII. Location for tests – Continued.

Title	Paragraph No.	Place of manufacture	Government proving ground
Boom check	4.5.1.43.1	X	X
Winch check	4.5.1.43.2	X	X
Winch brake test	4.5.1.43.3	X	X
Bulldozer test	4.5.1.43.4	X	X
Vehicle hydraulic system fluid leakage check	4.5.1.43.5	X	X
Control tests	4.5.2	X	X
Contamination check	4.5.2.3	X	X
Stowed equipment check	4.5.2.4	X	X
Fuel tank check	4.5.2.5	X	X
Vision devices and receptacle test	4.5.2.6	X	X
Turret control hydraulic fluid test	4.5.2.7	X	X
Boresight retention test, main sighting system	4.5.2.8	X	X
Boresight retention test, machine gun sighting system	4.5.2.9	X	X
Vehicle hydraulic system fluid test	4.5.2.10	X	X
Fire extinguisher control check	4.5.2.11	X	X
Gas particulate filter system	4.5.2.12	X	X
Comparison test	4.5.3		X
Cooling system test	4.5.3.2		X
Fuel system operational test	4.5.3.3		X
Ventilation test	4.5.3.4		X
Environmental tests	4.5.3.5		X
Speeds on grades check	4.5.3.6		X
Deep water fording test	4.5.3.7		X
Trench crossing test	4.5.3.8		X
Vertical obstacle test	4.5.3.9		X
Climbing test	4.5.3.10		X
Carbon dioxide distribution check	4.5.3.11		X

4.5.1.2 Shroud and sealing checks.

4.5.1.2.1 Sealing check. To determine conformance to 3.4.1.1 and 3.4.1.2 prior to and after fording operations, the vehicle shall be checked for leakage of water into crew compartment and the leaking or lubricants.

4.5.1.3 Hatch seal tests. To determine conformance to 3.4.1.3, the driver, loader, and commander hatches shall be closed and locked, and a spray of water directed on each hatch for a period of 3 minutes. All hatches shall be checked for leakage.

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4.5.1.4 Fuel system slope check. To determine conformance to 3.4.4, the vehicle shall be placed on specified grades and slopes, and engine operated and observed for functional requirements.

4.5.1.5 Fuel and hydraulic line check.

4.5.1.5.1 Fuel line check. To determine conformance to 3.4.4.2, the fuel system shall be pressurized with filtered air at a pressure of 3 to 4 psi prior to engine installation and checked for leaks at the fittings and fuel tank seals by applying soapy water or equivalent. The system shall be evacuated until a vacuum of 30 inches of water is attained.

4.5.1.5.2 Hydraulic system check. To determine conformance to 3.4.4.2, the hydraulic system shall be wiped clean and dry, system operated, and the lines and fittings checked for droplets (see 6.5).

4.5.1.6 Heater fuel feed check. To determine conformance to 3.4.4.4 with pressure inspection equipment installed at the heater, and all pumps operating as specified, fuel pressure at the personnel heaters shall be as specified.

4.5.1.7 Control checks. To determine conformance to 3.4.6, all controls shall be operated, adjusted, and checked for functional requirements.

4.5.1.8 Adjustment mechanism check. To determine conformance to 3.4.7, all adjustment mechanisms shall be checked for proper adjustment.

4.5.1.9 Driver's hatch and escape hatch check. To determine conformance to 3.4.7.1, force inspection equipment shall be attached to the driver hatch when it is in the closed position. The hatch shall be pulled to open position and back to closed position. The force shall be as specified. With the hatch in a closed position, the force required to lock the driver hatch cover shall be measured by inspection equipment at the end of the locking lever and shall be as specified. The driver escape hatch shall be checked by attaching force inspection equipment to the handle at point specified, releasing the hatch, and noting the force required. Caution shall be exercised in making the escape batch check by assuring that the area under the hatch is free of personnel and equipment before releasing the handle.

4.5.1.10 Loader and commander hatch check. To determine conformance to 3.4.8.5, attach a force gage to the locking latch and measure the force required to unlock the latch. Measure force required to push each hatch to its extreme open latch position. Close hatch and measure the downward force required to pull the hatch into a locking position and the rotational force required to lock the hatch.

4.5.1.11 Battery door check. To determine conformance to 3.4.7.2, force inspection equipment shall be attached to the battery door latch and the force required to open the latch shall not exceed that specified.

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4.5.1.12 Seating check. To determine conformance to 3.4.8 through 3.4.8.4, all seat levers shall be operated and the seats checked throughout their entire functional range. The force required to operate each lever shall be measured with inspection equipment and shall not exceed that specified.

4.5.1.13 Light checks. To determine conformance to 3.4.9.1 and 3.4.9.2, all lights shall be operated during and after the break-in run throughout all stages of operation, and checked for functional requirements.

4.5.1.14 Generator voltage check. To determine conformance to 3.4.9.7, the generator voltage shall be checked for functional requirements.

4.5.1.15 Generator blower motor check. To determine conformance to 3.4.9.8 and engine running at idle, the generator blower motor shall be operated and checked for constant flow of air.

4.5.1.16 Passive driver's night viewer power circuit check. To determine conformance to 3.4.9.9 with the master battery control and passive power switches turned on and vehicle generator supplying specified voltage, (see 3.4.9.7), the voltage output of the passive power pack shall be measured at the loose harness plug using an 80 megaohm load.

4.5.1.17 Slip ring and external communication circuit check. To determine conformance 3.4.9.4 and 3.4.9.15, the slip ring and external communication circuits shall be checked at all points required to assure continuity.

4.5.1.18 Auxiliary outlets. To determine conformance to 3.4.9.5, the OVE trouble light or suitable voltmeter shall be attached to the auxiliary outlets and checked for operation. The light or suitable voltmeter shall be removed and each circuit overloaded to check circuit breaker operation.

4.5.1.19 Searchlight control system check.

4.5.1.19.1 Vehicle and searchlight interface. To determine conformance to 3.4.9.6.1, the searchlight control circuits at the external connector shall be checked by inserting appropriate switching leads and jumper cables at control points. Certified fixtures shall be utilized to verify that mounting attachments for gun mount and control boxes have been installed to applicable drawings.

4.5.1.19.2 Searchlight installed. To determine conformance to 3.4.9.6.2, verify that the main gun has been balanced per 3.6.12.2 and align the gun on 600 meter or equivalent target. Elevate the gun 5 mils and perform the following:

a. With the main control box switch at "local" and beam switch at “compact", actuate the searchlight power switch to "visible" and verify that the searchlight beam can be centered on target.

b. Verify "spread beam" and "passive beam" functions.

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c. Place the power switch in the "off" position and actuate control switch to "remote" and repeat the applicable portions of a and b above.

d. At the conclusion of a through c above, visually verify that additional adjustment capability of the traverse and elevation adjusting screws exist.

4.5.1.20 Power plant electrical. To determine conformance to 3.4.9.10 during operation of the vehicle, the driver instrument panel shall be observed for proper display of engine functions as specified.

4.5.1.21 Air cleaner blower check. To determine conformance to 3.4.9.11 with the engine running at idle, air flow shall be detectable at each of the four air cleaner blower outlets. "No" flow at any of the outlets shall be cause for rejection.

4.5.1.22 Engine manifold heater check. To determine conformance to 3.4.9.12, the pushbutton switch on the purge pump handle and the starter switch shall be activated and pin "a" of the multiple connector of the engine electrical disconnect shall be checked for proper voltage.

4.5.1.23 Personnel heater. To determine conformance to 3.4.9.13, actuate the Hi-Lo switch to either position and verify reduced speed blower motor operation. Verify that the indicator lamp on the control panel lights within 4 minutes of turn-on as an indication of burner ignition. Subsequent to ignition, verify that the blower motor is operating at full speed and that high heat is generated when the Hi-Lo switch is in the Hi position and low heat is generated when in the Lo position. When the Hi-Lo switch is turned to the off position, verify that both the indicator lamp goes out and the blower motor shuts off within 3.5 minutes.

4.5.1.24 Engine smoke generator test. To determine conformance to 3.4.9.14 when the vehicle is equipped with an engine smoke generator, activate the smoke generator switch when the engine is warm and operating at 1600 rpm. Verify that the right and left exhaust ducts emit white smoke.

NOTE: Do not test the engine smoke generator for more than 15 seconds nor within a building or with personnel exposed to the exhaust plume.

4.5.1.25 Provisions for radio installation. To determine conformance to 3.4.9.15 (b) when a radio is not specified, an AN/VRC series radio set shall be installed and functionally inspected through the AN/VIC-1 (V) intercom set to insure proper operation.

4.5.1.26 Main gun firing circuits. To determine conformance to 3.4.9.16, the main gun firing circuit shall be checked using a special firing circuit tester which simulates the electrical firing action of a conventional round. Insert tester into the gun and close the breech. Each of the firing circuits (gunner power, manual elevation, and commander) and the auxiliary firing unit shall be actuated. The firing circuit tester shall be constructed to emit a discernable signal upon circuit completion. Absence of a discernable signal upon trigger or firing unit action, shall be cause for rejection

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4.5.1.27 Machine gun firing circuits check. To determine conformance to 3.4.9.17, the machine gun firing circuits shall be checked using inspection equipment. Each of the firing triggers should be qualified: gunner power control handle (2), manual elevation pump handle, commander handle, and cupola screw jack shall be checked for functional requirements. When machine gun firing triggers are actuated, listen for audible "click" of machine gun solenoid.

4.5.1.28 Main gun and machine gun circuit controls. To determine conformance to 3.4.9.18, commander controls and gunner controls shall be checked for ability to fire the weapons specified. The loader safety switch shall be closed in the fire position and open in the safe position to control the commander and gunner firing circuits to the main gun.

4.5.1.29 Smoke grenade discharger circuits.

4.5.1.29.1 Arming. To determine conformance to 3.4.9.19.1, operate the master battery switch to on, the turret power switch to on, and the grenade system power switch to on, and verify that the power-on (ready) lamp in the grenade power box illuminates. Operate the grenade system power switch to off and verify that the power-on (ready) lamp is extinguished.

4.5.1.29.2 Circuit voltage (complete installation). To determine conformance to 3.4.9.19.2, arm the circuit as specified in 4.5.1.34.1 and verify that the power-on (ready) lamp is illuminated. Actuate the right firing pushbutton and verify that the voltage on the smoke grenade discharger pins shown in figure 6 is as specified in 3.4.9.19.2. Actuate the left firing pushbutton and verify that the voltage on the smoke grenade discharger pins is as specified. Return the power-on (ready) switch to off and verify that voltage is not present on any of the firing pins.

4.5.1.29.3 Circuit voltage (Incomplete installation). When smoke grenade discharger and pushbutton unit are not installed on the vehicle, the external firing circuits at the external connector shall be checked on each vehicle by inserting appropriate switching leads at the internal control point. The complete installation shall be checked on a control basis in conjunction with stowed equipment (OVE) tests of 4.5.2.4.

4.5.1.30 Power gackage and train tests. To determine conformance to 3.6.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant power train, final drives, tracks, suspension system, and controls checked for functions.

4.5.1.31 Level road speed test. To determine conformance to 3.6.3.1, the vehicle shall be operated at maximum speed for not less than 10 minutes and at minimum speed for not less than 5 minutes and checked for function of components. Drift shall be not more than that specified. During the maximum speed test cycle, the vehicle shall maintain a speed of not less than 30 mph, 100 yard distance 10 consecutive times.

4.5.1.32 Acceleration test. To determine conformance to 3.6.4, the vehicle shall be operated as specified and the distance and time checked.

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4.5.1.33 Engine starting on grades and slopes test. To determine conformance to 3.5.2 and 3.6.5, the engine shall start and oil pressure and temperature shall be maintained when operating on specified grades and slopes with fuel tanks filled to no more than 50 percent of capacity. The 60 percent grade test may be conducted in conjunction with brake holding test.

4.5.1.34 Stopping test. To determine conformance to 3.6.6.1, the vehicle shall be operated on dry, level, hard-surfaced roads without loose material and the results of three consecutive stopping tests shall be averaged to make this determination.

4.5.1.35 Holding test. To determine conformance to 3.6.6.2, the added load required to simulate combat loaded vehicle shall be placed on the vehicle in position that will not restrict the flow of engine air (intake or exhaust). The vehicle shall be driven up a 60 percent dry, hard grade. Service brakes and parking brakes are checked for conformance. The engine starting and fuel system tests on a 60 percent slope may be conducted in conjunction with this test.

4.5.1.36 Turning test. To determine conformance to 3.6.7, vehicle shall be operated and pivoted 360° to right and left in neutral steer range and checked for functional requirements. During this test, the vehicle shall be brought to a full stop before reversing turning direction. The full pivots shall be accomplished within 35 feet diameter circle.

4.5.1.37 Fording test. To determine conformance to 3.6.8.1 and 3.6.8.5, the vehicle shall be driven into water of specified depth with engine compartment drain valves open and without special equipment. During or after test period water accumulation on crew compartment hull floor shall not be more than specified depth. The engine shall idle at 1,000 rpm for 15 minutes, stopped, and restarted in not more than 3 minutes with all accessories functioning satisfactorily during and after fording. The procedure employed in maintaining test facility cleanliness shall be furnished to the Government.

4.5.1.38 Drain valves check. To determine conformance to 3.6.8.3, inspection equipment shall be attached to the end of the handgrip of the front valve lever and actuated as specified. The rear valve lever shall have the gage attached and actuated as specified. The front and rear drain valve levers shall operate at forces specified and valve checked for functional requirements.

4.5.1.39 Inflatable seal and pump check. To determine conformance to 3.6.8.4, the pump shall be actuated to arrive at specified pressure as indicated on inflatable seal pressure gage. The indicated pressure loss during a 30 minute period shall not exceed 5 psi. Caution: the pressure shall be relieved at the conclusion of this test.

4.5.1.40 Turret and gun control system checks. To determine conformance to 3.6.12.1 with the turret level within 1°, the following tests shall be performed. Gun elevation and depression tests may be performed in conjunction with gun fire control tests. Throughout these

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tests, appropriate electrical checks may be made to assure adequate vehicular electrical power is being supplied and that the systems meet all the requirements at the voltage extremes specified. Magnetic brake must be de-energized before power traversing turret.

4.5.1.40.1 Main armament balance check. To determine conformance to 3.6.12.2, a gage shall be attached to the hand crank after turret and gun have been set to the conditions specified. The torque required to move the gun in either direction shall be measured and the gun shall balance as specified.

4.5.1.40.2 Nylon ballistic shield. To determine conformance to 3.6.12.3, the following tests shall be performed concurrently with synchronization tests of 4.5.1.41.1:

a. Coaxial machine gun deflection. With the main gun at boresight position point, a simulated machine gun containing a centering aperture shall be installed in the machine gun mount. The mount shall be adjusted to boresight position while viewing the coaxial machine gun target through the centering hole in the simulated machine gun. A deflection measuring device shall be mounted to the gun shield and coaxial machine gun deflection shall be checked at main gun elevation of 20° and minus-10°.

b. Gunner telescope sighting system deflection. With the gunner telescope set initially in its boresight position, deflection shall be tested by observing the telescope line of sight at 15° and minus 5° main gun elevation.

4.5.1.40.3 Manual traverse effort test. To determine conformance to 3.6.13.1 with the boom in erect position, the hand crank shall be rotated and torque to maintain turret movement shall be measured with inspection equipment. The mean torque shall be measured in each direction of rotation at four points throughout 360° of traverse.

4.5.l.40.4 Manual elevation effort test. To determine conformance to 3.6.13.2, the hand crank shall be rotated and torque to maintain movement of the gun in elevation and depression shall be measured with inspection equipment, with the gun at horizontal position plus or minus -1° at 15° or minus -1° elevation, and at minus -5° plus or minus –1° depression.

4.5.1.40.5 Manual elevation and depression response rate test. To determine conformance to 3.6.13.3, the hand elevation crank shall be rotated at the specified rate in elevation and depression. The gun angular movement per revolution shall be measured by means of the M13A3 quadrant.

4.5.1.40.6 System backlash. To determine conformance to 3.6.13.4, the backlash test shall be conducted using gunner telescope to measure the amount of backlash. A perpendicular force of 170 pounds shall be applied at the muzzle end of gun and exerted first in the traverse plane of rotation and then in the elevation plane. Backlash is the angular movement of the gun in traverse and elevation when the applied force is reversed. Before conducting this test, assure that the backlash has been adjusted per the installation drawing.

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4.5.1.40.7 Simultaneous use of power and manual control tests. To determine conformance to 3.6.13.5, the gun shall be elevated and depressed by power while traversing the turret with the manual controls. The gun shall be elevated and depressed by manual control while traversing the turret with power control. The elevation and traverse power switch shall be turned on and off not less than three times and the gun observed for movement.

4.5.1.40.8 Override control test. To determine conformance to 3.6.13.6, the override control switch shall be operated as specified and the control and movement of the gun observed. The commander override switch shall not be reactivated at intervals or less than 10 seconds.

4.5.1.40.9 Control system dead spot test. To determine conformance to 3.6.13.7, gunner and commander control handles shall be operated as specified and deflection of the handles to initiate movement of gun shall be measured with inspection equipment. When measuring the dead spot angle, the control handles shall not be moved toward the neutral center once the gun and turret have begun to move.

4.5.1.40.10 Gun elevation speed test. To determine conformance to 3.6.13.8, the gun shall be elevated and depressed at specified speeds by use of the gunner and commander power controls and the gun speed measured for maximum and minimum conformance. Observe the gun speed control at any rate between minimum and maximum speed.

4.5.1.40.11 Turret traversing speed test. To determine conformance to 3.6.13.9, by use of gunner and commander power controls, the turret shall be traversed clockwise and counterclockwise as specified, and minimum and maximum speeds shall be measured. Observe the turret speed control at any rate between minimum and maximum speed. The accumulator shall be fully charged prior to beginning the maximum speed test. Maximum traverse speed shall be determined by measuring time for the turret to complete 1 revolution from a standing point.

4.5.1.40.12 Main gun stability. To determine conformance to 3.6.13.10, the following test shall be performed with the vehicle nose down on a 15° slope, main gun horizontal, and 90° to the vehicle front to rear axis, turret power switch on, and engine inoperative:

a. Using the gunner telescope and manual elevation handle, aline the main gun on a target within 1° of horizontal.

b. Install a magnetic base dial indicator with a minimum 1 inch range on the turret roof and set it at a predetermined point on the gun breech.

c. Install a similar indicator on the race ring and set it at a predetermined point on the race ring gear.

d. Using the telescope and power control handle, depress and traverse the gun at least 15 mils off-target (without overloading either indicator) and then return the gun to the original target (a) without overshoot.

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e. Read both indicators within 8 seconds after arriving on target and again after a 1 hour period.

f. Verify that the gun has not moved more than 0.5 mil in elevation or traverse.

4.5.1.40.13 Elevation and traverse limit tests. To determine conformance to 3.6.13.11, the gun shall be operated in elevation, depression and traverse using power and manual controls individually and the angles measured in each of the defined areas.

4.5.1.40.14 Gun laying on stationary target test. To determine conformance to 3.6.13.12 for acceptance testing, the gunsight reticle shall be brought within the borders of the target specified for positions three and five with gunner power controls and position three with manual controls. Two trials shall be made and the maximum time shall be used for final determination. All positions shown on table III shall be used when performing CT's at the Government proving grounds.

4.5.1.40.15 Operation on slope test. To determine conformance to 3.6.13.13 for acceptance testing, the turret shall be placed on a 15° plus 2°, minus -1° slope and the gunsight reticle shall be brought within the borders of the target specified for positions three and five of table III. One determination shall be made for each position and direction specified using gunner power controls. Time limitation need not be met. All positions shown on table III shall be used when performing CT's at the Government proving grounds.

4.5.1.40.16 Turret locking test. To determine conformance to 3.6.13.14 with turret lock disengaged and the gun in forward position and elevated 3° to 8°, the vehicle shall be operated on smooth, level, hard-surfaced roads in low gear at speeds of 5 mph, then given a full steer 90° left turn and turret slippage noted on the azimuth indicator. This test shall be repeated with a full steer 90° right turn. The vehicle shall be driven on a straight course over the break-in run track at 20 mph, stopped as specified, and the gun slippage noted with the M1A1 gunner quadrant. The hand traverse crank and hand elevation crank shall be observed for conformance to specified requirements during the above tests.

4.5.l.40.17 Azimuth indicator test. To determine conformance to 3.6.16.2, the boresight deflection knob scale of the gunner periscope shall be set at four. The turret shall be traversed in a clockwise direction without over travel and vertical line of the gunner periscope reticle shall be laid on a fixed well defined vertical aiming point. The zero of the gunner aid (outer movable dial) on the azimuth indicator shall be positioned in line with the 1 mil pointer. The turret shall be traversed approximately 50 mils clockwise and returned counterclockwise to zero without over travel as read on the gunner azimuth indicator. The deflection boresight knob shall be rotated until the vertical line of the gunner periscope reticle is realined with the original aiming point. The difference between the original position and the final reading on the deflection boresight knob scale is backlash error and shall not exceed 0.5 mils.

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4.5.1.40.18 Turret control system fluid leakage check. To determine conformance to 3.6.13.18.1 and 3.6.13.18.2, the turret and gun control system components shall be wiped clean before performing tests. During or after 5 cycles of elevating the main gun in the interference zone, the leakage from the deck clearance valve shall not exceed that specified. During the testing specified in 4.5.1.40.14, the fluid leakage from the override linkage control box shall not exceed that specified.

4.5.1.40.19 Hydraulic pressure limits test. To determine conformance to 3.6.13.15, the turret shall be traversed as specified and the system pressure gage observed for specified pressure as the power pack motor comes on. With power pack motor running and the controls in their neutral position, the power pack motor shall cut out at the specified system pressure.

4.5.1.41 Gun fire control system tests. To determine conformance to fire control system requirements of 3.6.14, the following tests shall be performed. With the exception of boresight retention, these tests may be conducted on an equivalent indoor turret stand using equipment defined herein. The portion of the vehicle tested using this method shall be the turret and other installations as assembled just prior to marriage of hull and track portion of the vehicle. Turret input voltage shall be as specified in 3.6.12.1.

4.5.1.41.1 Synchronization tests. To determine conformance to 3.6.14.1 synchronization tests shall be accomplished using figure 1 through figure 4 and the following equipment:

a. Turret stand. The turret mounting surface of the turret stand shall be level within 0.2 mils.
b. Target and target stand. The targets (aiming points) shall be located on the target stand as shown in figure 3. Targets shall be in a common plane within 1/16-inch. Details of aiming points shall be as shown in figure 4. The plane formed by the zero gun aiming point 15° gun aiming point, and the 5° periscope aiming point shall be vertical within 0.5 mil. The distance from this plane to the centerline of the turret stand shall be 522.5 inches plus or minus 1/8-inch. A vertical plane perpendicular to the aforementioned plane and intersecting the zero gun aiming point shall intersect the zero gun aiming point shall intersect the centerline of the turret mounting diameter of the turret stand within 1/4-inch.
c. Alignment telescope, split level, and breech adapters the adapter shall locate the split level or the alignment telescope in the main gun breech.
d. Gunner periscope aperture gage.

4.5.1.41.1.1 Checks and adjustments. With turret assembly installed on stand and stand level as specified, the gun, sighting instruments, and test equipment shall be adjusted as follows:

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a. Alignment telescope and breech adapter shall be inserted in gun tube at breech end of main gun. The alignment telescope shall be focused on target board for sharp definition and minimum of parallax as viewed from the breech position. The gun shall be laid on its zero elevation point.
b. The gun tube shall be leveled by laying gun from low to high until the split level inserted in the barrel adapter reads level.
c. Target shall be positioned with respect to gun trunnions as follows:

    (1) When the gun is leveled in (b) above, the horizontal (zero) gun aiming point target shall be in line with the crosshair reticle as viewed through the alignment telescope within 1/8-inch. The target board shall be raised or lowered to position the aiming point.
    (2) The gun shall be elevated to its 15° aiming point target and the target board rotated, if necessary, to superimpose the reticle on the 15° aiming point. When the target board is positioned so that elevation or depression of the gun results in the reticle traveling exactly through the zero elevation and 15° aiming points, the target shall be within 2 mils clockwise or counterclockwise of a plumb line position.
    (3) Laying from low to high, the alignment telescope reticle shall be superimposed on the zero elevation aiming points.

d. Aperture gage shall be placed over the objective window of the gunner periscope.
e. Reticles of the gunner periscope and telescope shall be alined with corresponding aiming points by rotating the azimuth and elevation boresight knobs.

4.5.1.41.1.2 Procedure. The alignment telescope reticle shall be superimposed on each of the aiming points laying low to high without over travel. The turret shall be traversed if necessary. At each position above horizontal, the gunner periscope cross hairs (daylight system) shall fall within its respective tolerance bands if half or more of the reticle line width is in the white area.

4.5.1.41.2 Sighting system backlash. To determine conformance to 3.6.14.2, backlash shall be checked after synchronization requirements have been met. Tests shall be made in accordance with the following procedure:

a. Gun shall be placed "dead-on" aiming point of target by laying from low to high. This shall be accomplished without over travel.
b. The gunner periscope daylight systems aiming reticle shall be placed "dead-on" target relative to the gun. Boresight knobs shall be locked in this position and micrometer scales slipped to their normal settings three or four.
c. The gun shall be realigned with the same target aiming point as in a by laying from high to low without over travel.

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d. If necessary, sight reticle (excluding boresight knob backlash) shall be realigned "dead-on" aiming point by use of elevation boresight knobs. Elevation boresight knob micrometer scale shall be used.
e. The difference in reading between b and d above shall not exceed the tolerance specified.
f. Backlash shall be checked at gun positions of approximately 0°, 5°, 10°, and 15° elevations.

4.5.1.41.3 Boresight knob travel test. To determine conformance to 3.6.14.3 after confirming that synchronization requirements have been met, boresight knob travel shall be tested from the positions established in 4.5.1.41.1.1.

4.5.1.41.4 Leveling device tests. To determine conformance to 3.6.16.3, the gun shall be leveled at 0° elevation as indicated by the level on the alignment telescope adaptor positioned in the gun chamber. The leveling device on the ballistic drive shall be observed for accuracy.

4.5.1.41.5 Elevation quadrant M13A3 test. To determine conformance to 3.6.16.1, with the turret in a level position using the M1A1 calibrated gunner quadrant on the breech pads, the gun shall be elevated on 355 mils. The reading on the elevation quadrant M13A3 shall not exceed that specified. The procedure shall be repeated with the gun at a depression of 124 mils.

4.5.1.42 Cupola test.

4.5.1.42.1 Cupola slip ring contact board test. To determine conformance to 3.6.15.1, the slip ring contact boards shall be checked for vertical adjustment with their corresponding contact spring.

4.5.1.42.2 Cupola slip ring circuit test. To determine conformance to 3.6.15.2, the current transfer between turret and cupola shall be checked at each of the slip rings. Current as specified, supplied by the power source, shall be delivered through the slip rings, contacts, and voltage drop measured. An equivalent electrical load shall be connected in the cupola. The voltage drop when measured from the turret junction box to the cupola output power connector and cupola communication receptacle shall not exceed 2 volts during 360° traverse. The individual slip rings may be wired in series to conduct this check but the voltage drop shall not exceed a multiple of 2 volts depending on number of circuits simultaneously tested.

4.5.1.42.3 Cupola noise at slip rings. To determine conformance to 3.6.15.3, an audio generator shall be connected to the turret junction box with one side of the output connected to slip ring ground circuit. A 100 cycles per second (cps) signal shall be introduced into each ring, and cupola traversed 360° checking for static in the earphones at the cupola. The individual slip ring circuits may be wired in series for this check. The communications circuits shall be as specified.

4.5.1.42.4 Cupola traverse and elevation effort. To determine conformance to 3.6.15.4 through 3.6.15.6 each cupola azimuth gear box handwheel and screwjack handle shall be operated and checked for starting torque.

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4.5.1.42 5 Cupola interlock assembly location test. To determine conformance to 3.6.17.5, each vehicle shall be tested concurrently with synchronization tests of 4.5.l.41.1. With the main gun on its 0° aiming point as in 4.5.1.41.1.1 (c(3)), the cupola interlock shall be engaged and the alignment of the machine gun line of eight shall be checked using telescope in an adapter to view the machine gun target on the target stand (figure 3). The vertical cross line of telescope shall fall within the tolerance bands of the target.

4.5.l.42.6 Cupola machine gun sighting tests. To determine conformance to machine gun sighting system requirements specified herein, the cupola shall be mounted on a cupola mounting stand and tested prior to installation on the turret using equipment defined herein.

a. Cupola mounting stands.
b. Synchronization fixture (with M39 target mirror perpendicular to M/G line of sight).
c. Telescope (M/G line of sight).
d. Telescope Adapter (M/G mount and synchronizer fixture).
e. Reticle projector (beam splitter and beam splitter adaptor).
f. Collimator Fixture.
g. Collimator (M/G line of sight).
h. Collimator (2 req'd) periscope line of sight.

4.5.1.42.6.1 Cupola machine gun synchronization. To determine conformance to 3.6.17.1, synchronization tests shall be performed according to the following procedure:

a. The cupola shall be mounted on the cupola stand and the telescope adaptor shall be inserted in the machine gun mount.
b. The synchronization fixture shall be attached to the telescope adaptor.
c. One of the periscope heads shall be installed in its mount on the cupola.
d. The periscope body shall be inserted in the periscope head.
e. The reticle projector shall be attached to the periscope eyepiece.
f. Approximate 0° gun position shall be established, laying from low to high.
g. The boresight knobs of the periscope shall be adjusted to autocollimate the periscope on the mirror of the synchronization fixture and the boresight knob position shall be recorded.
h. Laying from low to high, the periscope shall be within the synchronization limits specified in 3.6.17.1.

NOTE: Synchronization error at any gun position shall be measured by noting the boresight knob movement necessary to superimpose the periscope reticle on its reflected image from the mirror of the synchronization fixture.

i. Steps d thru h shall be repeated with the other periscope head installed in the cupola.

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4.5.1.42.6.2 Backlash. To determine conformance to 3.6.17.2, backlash shall be checked after confirming that synchronization requirements have been met. Tests shall be made in accordance with the following procedure:

a. Approximate 0° gun position shall be established laying from low to high.
b. The boresight knobs of the periscope shall be adjusted to autocollimate the periscope on the mirror of the synchronization fixture. The boresight knob position shall be recorded.
c. Approximate 0° gun position shall be established laying from high to low.
d. The boresight knobs shall be moved, if necessary, to superimpose the periscope reticle on its reflected image from the mirror of the synchronization fixture. Boresight knob movement from positions established in "b" above, shall not exceed the backlash limits established in 3.6.17.2.
e. Backlash shall be checked at gun positions of approximately 0°, 15°, 30°, and 50° elevation and 10° depression.

4.5.1.42.6.3 Boresight knob travel. To determine conformance to 3.6.17.3, boresight knob travel shall be tested after synchronization requirements have been met. Tests shall be performed according to the following procedure:

a. With the periscope removed from its mount, the synchronization fixture shall be removed from the telescope adaptor and the telescope shall be inserted in the telescope adaptor.
b. The telescope shall be aligned with its collimator of the collimator fixture laying from low to high without over travel.

NOTE: The machine gun centerline shall not be disturbed for remainder of boresight knob travel inspection steps.

c. The telescope shall be removed from its adaptor.
d. One periscope body and one periscope head shall be installed in the periscope mount.
e. The boresight knobs shall be adjusted to superimpose the periscope reticle on its collimator. Boresight knob travel from this position shall be as specified in 3.6.17.3.
f. The installed periscope body shall be replaced by the other periscope body and step (e) above shall be repeated using the respective collimators for the periscope body.
g. Steps e and f above shall be repeated with the other periscope head in the periscope mount.

4.5.1.43 Vehicle hydraulic control system tests.

4.5.1.43.1 Boom check. To determine conformance to 3.6.18.1 and 3.6.18.2 the boom travel locks shall be disengaged. The winch cable shall be over the boom tip sheave and fastened to the hook stowage eye. The winch gear shift shall be in low gear. To raise the boom from

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stowed position to vertical position, the boom control valve shall be displaced fully in "raise" position until the boom is vertical position. The winch control valve shall be held in "reel-in" position until the boom has been pulled back to approximately 6 inches above turret travel locks and stowed travel locks by metering boom control valve.

4.5.1.43.2 Winch check. To determine conformance to 3.6.18.3, place the boom in erect position against stayline and with the winch in "LO" gear attach the boom cable hook to a weight of not less than 187,500 pounds, raise the weight 1 foot and suspend it for one minute with the control valve in "Neutral", then continue raising the weight to a point 2 feet above the initial starting position, stop, and lower to the ground. With the cable attached to a weight of not more than 20,000 pounds, the relief valve setting shall prevent it from being lifted. With sufficient tension to maintain proper reeling and with the engine operating as specified, the rate of reel-in shall be checked for feet per minute.

4.5.1.43.3 Winch brake check. To determine conformance to 3.6.18.4 with weight specified attached to cable, the winch shall be operated and the brake checked for functional requirements.

4.5.l.43.4 Bulldozer test. To determine conformance to 3.6.18.5, the bulldozer control shall be operated as specified and operation of bulldozer observed for functional requirements.

4.5.1.43.5 Vehicle hydraulic system fluid leakage check. To determine conformance to 3.6.18.7, the vehicle hydraulic system components shall be wiped clean prior to testing. The hydraulic system shall be operated through all operations and checked for operational requirements.

4.5.2 Control tests. Control tests for maintaining control of those requirments whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein.

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first 10 vehicles produced, and thereafter one per month or as specified, shall be checked in accordance with the following control tests.

4.5.2.2 Failure. If the vehicle selected fails to pass any control tests, the Government inspector shall stop acceptance until such time as conditions causing the failure have been remedied.

4.5.2.3 Contamination check. To determine conformance to 3.6.8.5, the vehicle after the water fording test, shall be examined for contamination of lubricants of the transmission, engine, final drive, reservoir, and road wheel two, three, four, five, and one of their corresponding arms on each side of the vehicle. Contamination shall not exceed 2 percent by volume.

4.5.2.4 Stowed equipment check. To determine conformance to 3.4.2, all manufacturer and depot installed OVE shall be stowed (combat loaded) on the selected vehicle. All items shall fit in the space provided. The equipment shall be removed after the test. The OVE used shall be of the latest configuration available to the contractor. The contractor shall be permitted to unpack and repackage equipment used for test.

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4.5.2.5 Fuel tank check. To determine conformance to 3.4.4.1, the fuel tank shall be filled at rate of not less than 50 gallons per minute and observed for leaks before and after a road test on smooth, level, and hard-surface road.

4.5.2.6 Vision devices and receptacle tests. To determine conformance to 3.4.10, the gunner and commander periscopes shall be installed in the proper receptacles and checked for line of sight. Clamp, observe deviation, record, and unclamp through four complete cycles. The four determinations shall be averaged and the resulting value assigned as the line of sight deviation for the instrument. All vision devices and receptacles, except articulating telescope, shall have a spray of water directed on each for a period of 3 minutes. All devices shall be checked for leakage in conjunction with 4.5.1.2.1.

4.5.2.7 Turret control hydraulic fluid test. To determine conformance to 3.6.13.18.1 and 3.6.13.18.2, the hydraulic oil in the control power pack shall be tested for water contamination and cleanliness and leakage.

4.5.2.8 Boresight retention test, main gun sighting system. To determine conformance to 3.6.14.5, the following procedure shall be followed:

a. Prior to 8 mile road test (see 4.5.1.1), vehicle shall be positioned at least 600 meters from clearly defined target.
b. Gun shall be placed "dead on" target by laying from low to high without over travel.
c. All sighting equipment shall be placed "dead on" same target chosen in b. Boresight knobs shall be locked in this position and micrometer scales slipped to their nominal position (3) or (4).
d. Eight mile road test shall be performed.
e. Vehicle shall be returned to position established in a.
f. Step b shall be repeated.
g. The boresight knobs (excluding boresight knob backlash) of gunner periscope shall be used to position its reticle "dead on" target and the error determined from boresight knob scale readings.
h. With the gunner daylight system periscope boresighted on target, the reticle of the gunner M105F telescope shall be laid on the same target using the gun elevating and turret traversing controls.
i. The gunner periscope daylight system reticle shall be laid on target with the periscope boresight knobs noting the mil travel on the knobs.
j. The reading obtained in i shall be the shift in boresight of the gunner M105F telescope.
k. Through this operation, the gunner M105F telescope shall remain in locked position. The boresight knobs on the gunner telescope mount shall not be used for checking shift in boresight.

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If any boresight knob has to be moved more than values specified to bring the line of eight of an instrument "dead on" target, one more determination of retention shall be made after another 8 miles of operation. All instruments shall maintain boresight.

4.5.2.9 Boresight retention test, machine gun sighting system. To determine conformance to 3.6.17.4, the following procedure shall be followed:

a. Prior to 8 mile road test (see 4.5.1.1), the vehicle shall be positioned at least 600 meters from clearly defined target.
b. Gun shall be placed "dead on" target by laying from low to high without over travel.
c. The periscope reticle shall be placed "dead on" same target chosen in b. Boresight knobs shall be locked in this position and micrometer scales slipped to their nominal position.
d. Eight mile road test shall be performed.
e. Vehicle shall be returned to position established in a.
f. Step b shall be repeated.
g. The boresight knobs, excluding boresight, knob backlash, of the periscope shall be used if necessary to position its reticle "dead on" target and the error determined from boresight knob scale reading. Boresight retention shall be within the tolerance specified in 3.6.17.4.

If either boresight knob has to be moved more than the value specified to bring the line of sight of the periscope "dead on" target, one more determination of retention shall be made after another 8 miles of operation and the periscope shall maintain boresight.

4.5.2.10 Vehicle hydraulic system fluid test. To determine conformance to 3.6.18.6 the hydraulic oil in the vehicle control system shall be tested for water contamination and cleanliness.

4.5.2.11 Fire extinguisher control check. To determine conformance to 3.8, one vehicle per month shall have the fire extinguisher test for operation. With the engine idling at 700 to 800 rpm, the internal control shall be pulled with a maximum effort of 55 pounds using a force gage. The first shot shall actuate one cylinder. After a time delay of not more than 11 seconds, the engine shall have stopped and the system shall begin discharging CO_2 into the engine compartment and in the center of the engine “V”. After returning the handle to its original position, a second actuation shall discharge the remaining two cylinders of CO_2. Removal and weighing of each CO_2 cylinder after test shall show a minimum of 10 pounds CO_2 discharged during test (see 6.3).

4.5.2.12 Gas particulate filter system test. To determine conformance to 3.9.1 and 3.9.2, the existing ambient temperature shall be recorded and all crew outlets shall be opened. The air flow at each outlet shall be checked and shall be as specified. When the temperature has risen 10° and proceeding conditions have been specified, the heater shall be considered properly installed.

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4.5.3 Comparison tests. The Government may select vehicles any time during the contract production period and subject vehicles to all tests as listed on table VIII and table IX to reveal deficiencies of manufacture or materials that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Comparison test vehicles shall be combat loaded with all OVE stowed in their applicable spaces to conform to 3.4.2. Vehicles selected shall not include any vehicles previously tested for conformance to 4.5.2.

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

TABLE IX. 2,000 Mile test combat loaded (Government Proving Ground).

Course	Mileage and speeds
Hard surface roads	200 miles at varying speeds up to maximum.
Gravel and dirt roads	500 miles at varying speeds up to maximum.
Level cross country	650 miles at varying speeds up to maximum.
Hilly cross country	650 miles at varying speeds up to maximum.

4.5.3.2 Cooling system tests. To determine conformance to 3.4.3 with the vehicle operating in specified temperature, the cooling system shall be checked for functional requirments (see 4.5.3.5).

4.5.3.3 Fuel system operational test. To determine conformance to 3.4.4, the vehicle shall be operated through all speed and gear ranges over cross country, hilly terrain, longitudinal grades, lateral slopes, and on smooth, level hard-surface roads, and vehicle observed for functional requirements.

4.5.3.4 Ventilation test. To determine conformance to 3.4.5, the vehicle shall be "buttoned up" and the engine operated at 900 to 1000 rpm for a period of not less than 30 minutes. With all ventilating systems in operation, the main gun and machine gun shall be fired as follows:

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a. Main gun firing contamination levels shall be determined for separate three round salvos for each type of ammunition designated for the main weapon.

b. Secondary weapon contamination levels shall be determined for one 200-round burst fired from each vehicle machine gun simultaneously.

4.5.3.5 Environmental test. To determine conformance to 3.6.1, vehicle properly serviced and equipped, shall be subjected to the high temperature specified and observed for performance and heating conditions. With vehicle operating in ambient air temperature of 115°F and a cooling air intake grille restriction not exceeding 2 inches H_2O, measured between cylinders with cooling air exhaust restriction but not exceeding 4 inches H_2O measured above the cooling fans for all transmission gear ranges above the 0.3 maximum speed ratio point, the vehicle shall meet the performance requirements. The vehicle properly serviced and equipped shall be subjected to the sub-zero temperatures and checked for operational requirments.

4.5.3.6 Speed on grades check. To determine conformance to 3.6.3.2, the vehicle shall be operated on specified grades at specified speeds and checked for operational requirments.

4.5.3.7 Deep water fording test. To determine conformance to 3.6.8.2, the vehicle with fording kit installed, shall be driven into water and checked for accumulation of water and tests performed to determine the extent of contamination in the lubricants (see 4.5.2.3). The vehicle shall be operated in specified depths of water for specified time periods and observed for function of engine and all other components.

4.5.3.8 Trench crossing tests. Vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over trenches as specified in 3.6.9. The boom shall be in the stowed position and securely clamped in the turret and hull travel locks. After test, the vehicle shall be examined for damage.

4.5.3.9 Vertical obstacle test. Vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over vertical obstacles as specified in 3.6.10. The boom shall be in the stowed position and securely clamped in the turret travel locks. After test, the vehicle shall be examined for damage.

4.5.3.10 Climbing test. To determine conformance to 3.6.11, each vehicle shall be driven in forward and reverse gear up an approved 60 percent grade and on 30 percent side slopes.

4.5.3.11 Carbon dioxide distribution check. To determine conformance to 3.9 and 4.5.2.11, the fire extinguisher system shall be tested and the results examined for adequate distribution and control. With the engine idling at 700 to 800 RPM and CO_2 concentration meters located at the approximate center of each perforated distribution tube, the internal controls shall be operated as specified, using Force Gage 8752278. The first shot shall actuate

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one bottle of CO_2. After a time delay of not more than 11 seconds, the system shall begin discharging CO_2 into the engine compartment and the center of the engine "V". After returning the handle to its original position, a second actuation shall discharge the remaining two bottles of CO_2. The CO_2 meters shall register a peak CO_2 concentration of at least 70 percent diminishing to zero within 1 minute after discharge. Removal and weighing of the CO_2 cylinders shall show a minimum of 10 pounds of CO_2 discharged.

4.5.3.12 Electromagnetic characteristics. To determine conformance to 3.4.9.4 and 3.10 through 3.10.4, the production vehicle test shall include electromagnetic interference tests in accordance with MIL-STD-462. As an aid to determining intra-system compatibility, broadband conducted emissions on the +28 vdc bus shall be compared to the limits of CE07. Broadband radiated emission limts shall be measured at a 6.0 +0.1 meter distance from the vehicle. Radiated susceptibility test RS03 shall be performed using 1.0KHZ sine wave modulation at 50 percent amplitude modulation as well as CW.

4.5.3.13 Vibration and noise level. To determine conformance to 3.11.1 and 3.11.2, the average of three noise level readings at each octave band shall be determine at each operator head position under normal operating procedures. The soundlevel meter, microphone-related equipment, and techniques employed shall be in accordance with MIL-STD-1474. Verify that noise hazard signs are clearly visible to personnel in all crew stations.

4.5.4 Preparation for delivery inspection. Preparation for delivery inspection shall be performed on each vehicle that has completed acceptance and control tests per the applicable requirements of MIL-V-62060. Final equipment and component installation shall also be inspected and tested, as required, to determine conformance to applicable requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with MIL-V-62060.

6. NOTES

6.1 Intended use. The vehicle is intended for use by the Armed Forces as a combat vehicle.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Preproduction sample if required (see 3.1).
c. Deep water kit installation when required (see 3.6.8.2).
d. Radio interference if other than specified (see 3.7).
e. Selection of applicable levels of preservation, packing and packaging (see 5.1).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

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6.4 Fuels and lubricants. Fuels and lubricants used for vehicle break-in run and acceptance tests shall be changed as necessitated by seasonal temperature changes and as directed by the contracting officer.

6.5 The following shall be used with the classification of defects for leak:

a. Weep: Any evidence of fluid beyond seal.
b. Seep: Any evidence of fluid beyond seal that does not result in formation of a droplet.
c. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: Any evidence of fluid beyond the seal where droplet forms and fall.
e. Drop: A drop is defined as a volume of 0.05 cubic centimeters.

6.6 International standardization agreement. Certain provisions (see 3.6.8.1) of this specification are the subject of international standardization agreement NATO STANAG 2805. When amendment, revision or cancellation of this specification is proposed that will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Changes from previous issue. Asterisks, to identify changes with respect to the previous issue, are not used in this revision due to the extensiveness of the changes.

Custodian:
Army – AT

Preparing activity:
Army – AT

Project No. 2350-A291

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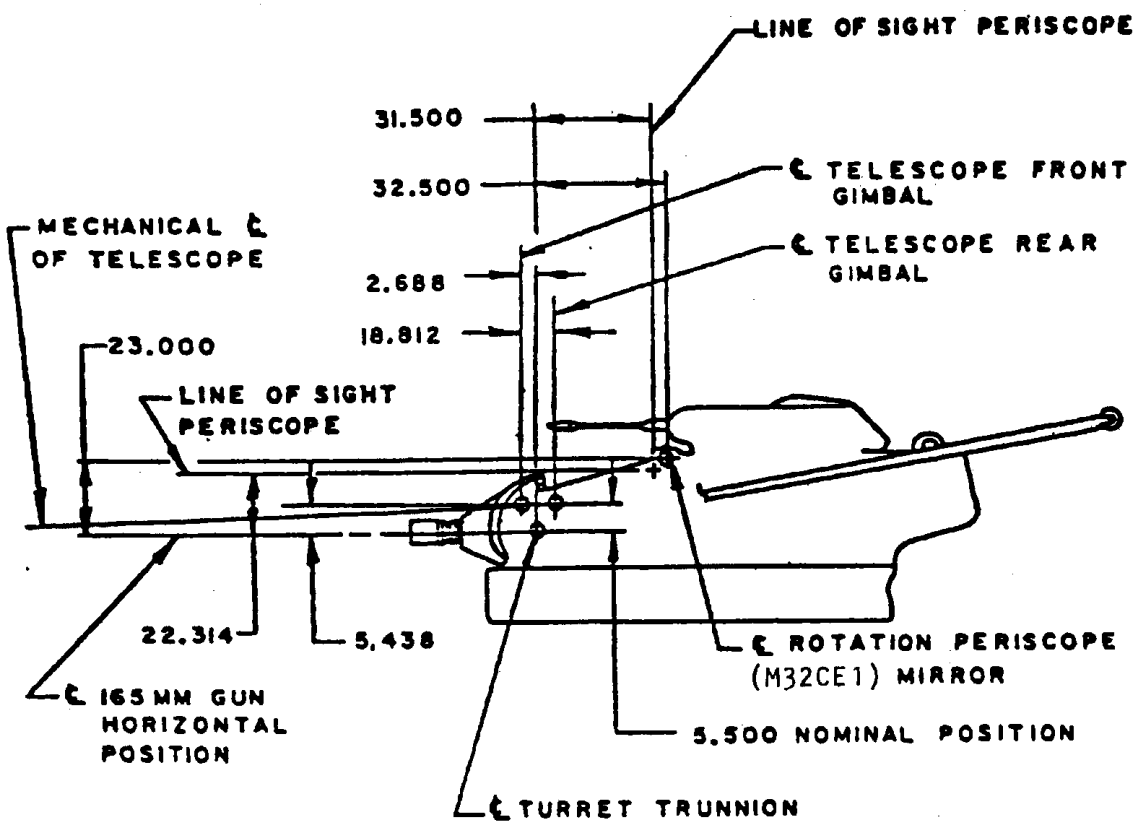


FIGURE 1. Side elevation.

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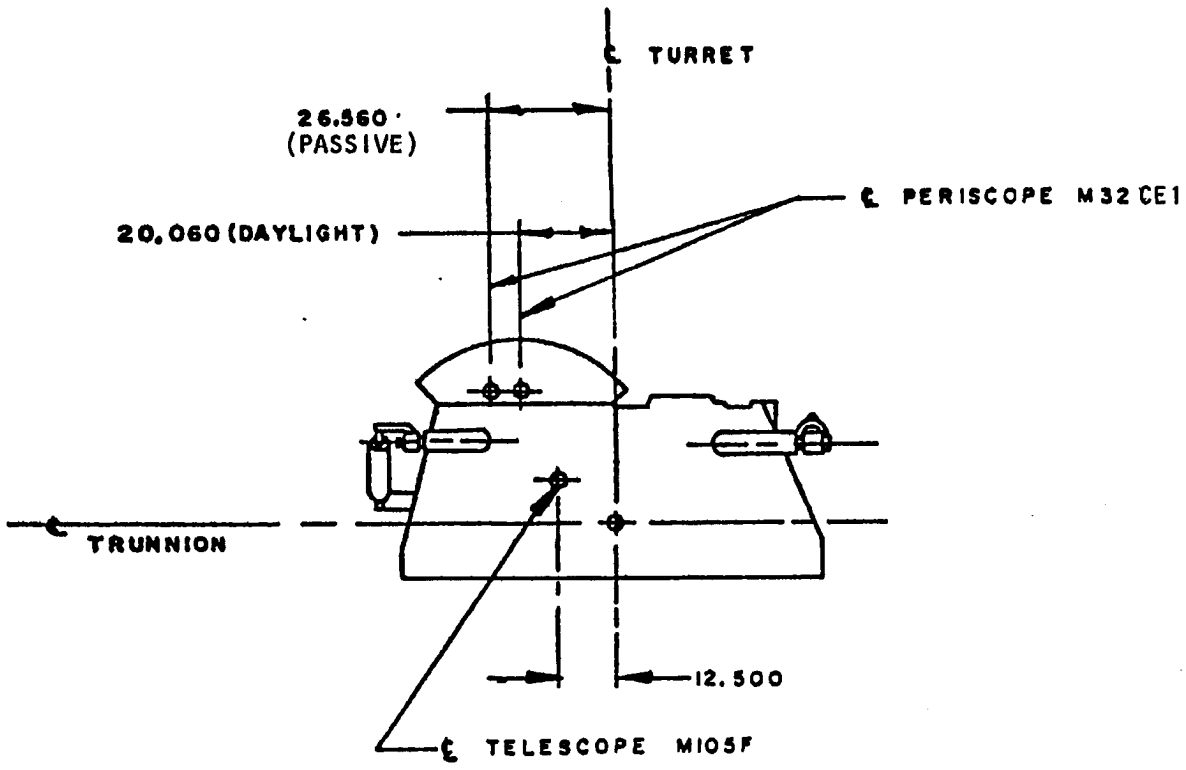
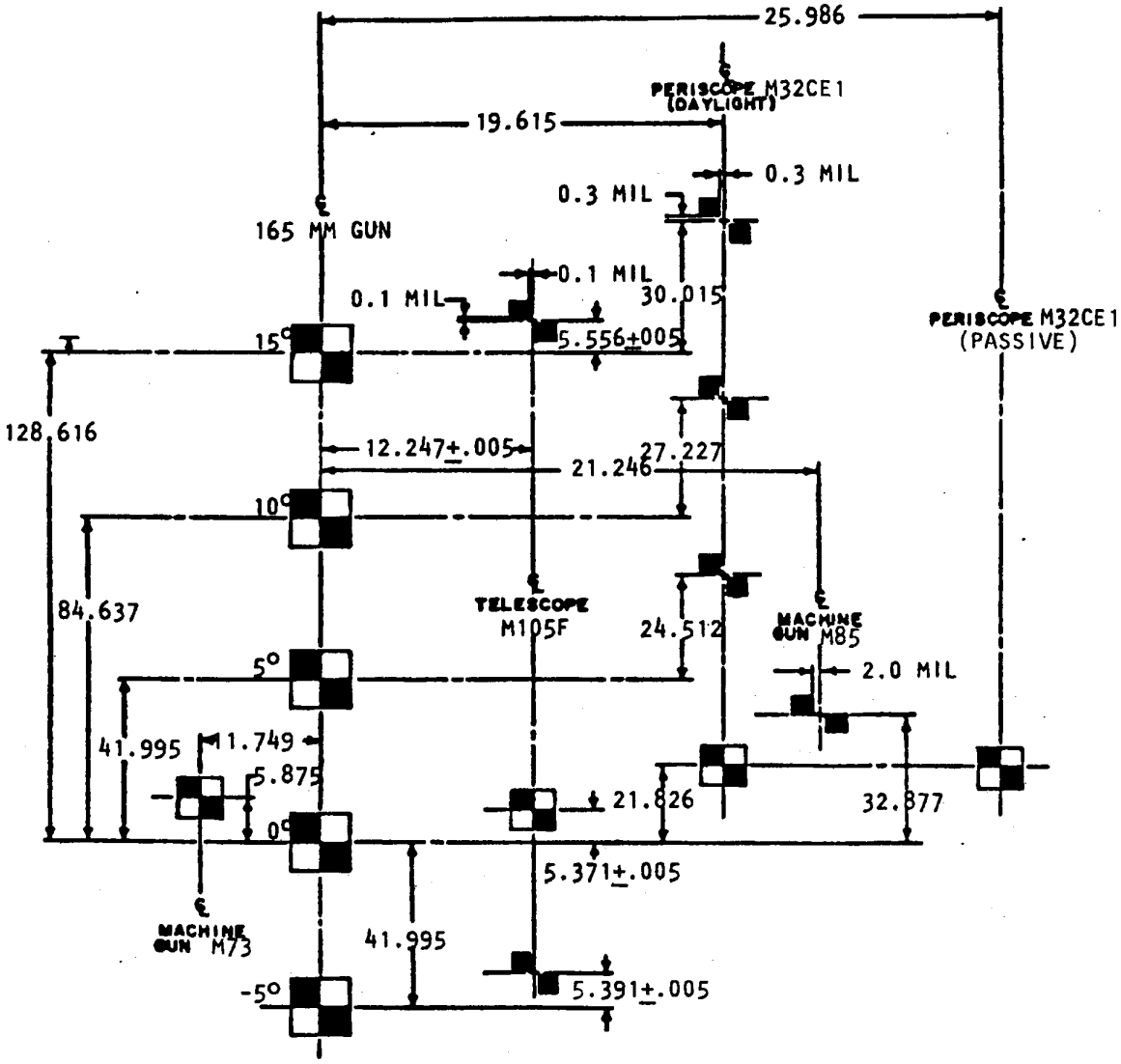


FIGURE 2. Front elevation.

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DISTANCE FROM TARGET TO ℄ GUN TRUNNIONS = 40 FEET

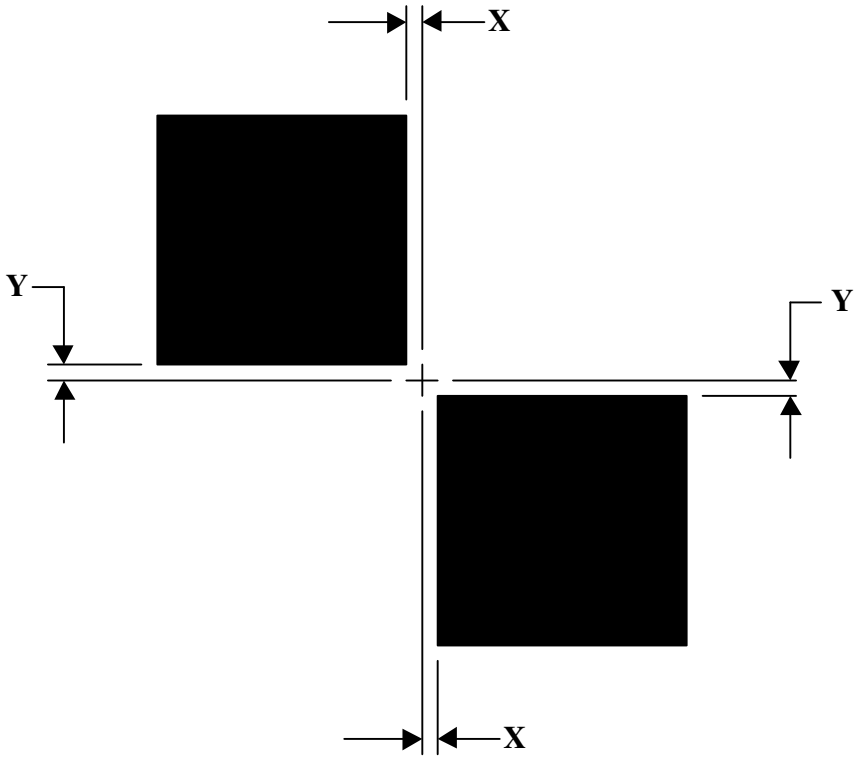
±.015 TOLERANCE ON ALL DIMENSIONS UNLESS OTHERWISE SPECIFIED. (ALL DIMENSIONS IN INCHES UNLESS OTHERWISE STATED.)

TARGET AIMING POINTS ARE POSITIONED TO REFLECT CONVERGENCE OF THE LINES OF SIGHT WITH THE MAIN GUN AT 600 METERS FOR ALL GUN POSITIONS.

FIGURE 3. Boresight and synchronization target.

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OPEN AIMING POINTS



	TOLERANCE BANDS OF TARGETS AT • 40 FEET BASED ON 600 METER BORESIGHT									
	GUN ELEVATION, DEGREES									
	-5		0		+5		+10		+15	
	Mil	Inches	Mil	Inches	Mil	Inches	Mil	Inches	Mil	Inches
Periscope M32 CE1										
X					0.3	.151	0.3	.153	0.3	.156
Y					0.3	.152	0.3	.156	0.3	.162
Telescope M105F										
X	0.1	0.48							0.1	.049
Y	0.1	0.48							0.1	.051
Machine Gun M85										
X			2.0	1.077						
Y			0.0	0.000						

FIGURE 4.

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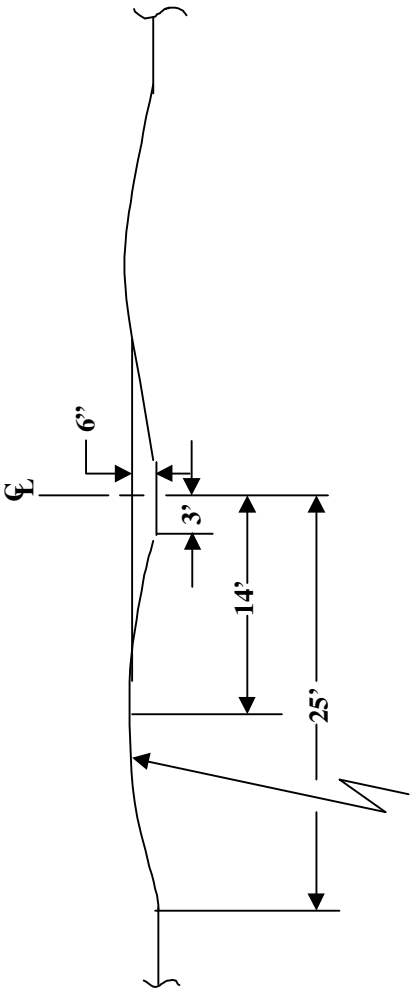


FIGURE 5. Profile of test track artificial bump.

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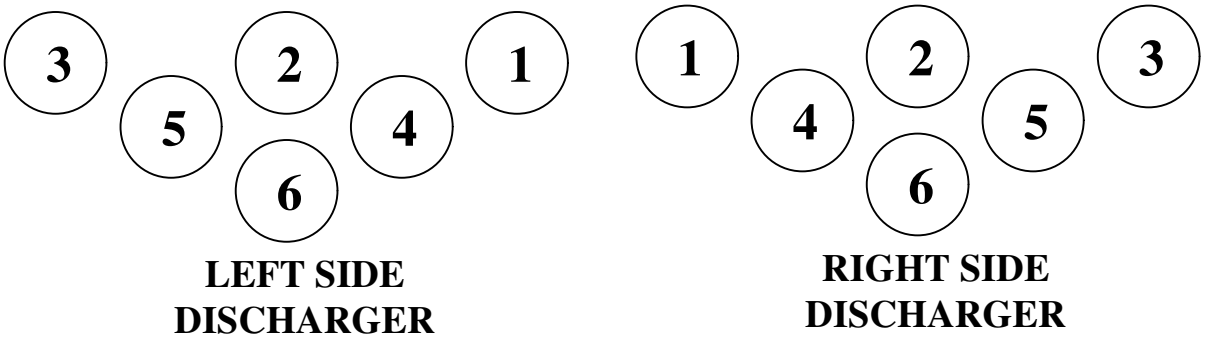


FIGURE 6. Grenade discharge barrel numbering.

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